



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

BE – CV

III to VIII Semester

**Jun/Jul-2024
2018 & 2022 Scheme**

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Bannur road, Alanahalli, Mysore-570028

Sl.No.	Sub-Code	Subject Title	Exam Date
1	18CV32	Strength of Materials	June/July 2024
2	18CV35	Basic Surveying	June/July 2024
3	18CV43	Applied Hydraulics	June/July 2024
4	18CV52	Analysis of Indeterminate Structures	June/July 2024
5	18CV54	Basic Geo Technical Engineering	June/July 2024
6	18CV55	Municipal Waste Water Engineering	June/July 2024
7	18CV56	Highway Engineering	June/July 2024
8	18CV61	Design of Steel Structural Elements	June/July 2024
9	18CV62	Applied Geotechnical Engineering	June/July 2024
10	18CV 63	Hydrology and Irrigation Engineering	June/July 2024
11	18CV643	Alternate Building Materials	June/July 2024
12	18CV72	Design of RCC and Steel Structures	June/July 2024
13	18CV734	Ground Water Hydraulics	June/July 2024
14	18CV745	Urban Transport Planning	June/July 2024
15	18CV81	Design of Pre -Stressed Concrete	June/July 2024
16	18CV824	Rehabilitation and Retrofitting	June/July 2024
17	21CV32	Geodetic Engineering	June/July 2024
18	21CV33	Strength of Materials	June/July 2024
19	21CV42	Fluid Mechanics and Hydraulics	June/July 2024
20	21CV44	Analysis of Structures	June/July 2024
21	21CV485	Green Buildings	June/July 2024

22	21CV51	Hydrology and Water Resources Engineering	June/July 2024
23	21CV52	Transportation Engineering	June/July 2024
24	21CV53	Design of RC Structural Elements	June/July 2024
25	21CV54	Geotechnical Engineering	June/July 2024
26	21CV61	Construction Management and Entrepreneurship	June/July 2024
27	21CV62	Concrete Technology	June/July 2024
28	21CV63	Design of Steel Structure	June/July 2024
29	21CV646	Alternative Building Materials	June/July 2024
30	BESCK104A/BESCKA104	Introduction to Civil Engineering	June/July 2024
31	BETCK105B/BETCB105	Green Buildings	June/July 2024
32	BBOK407	Biology for Engineers	June/July 2024
33	BCIVC103/203	Engineering Mechanics	June/July 2024
34	BCV301	Strength of Materials	June/July 2024
35	BCV302	Engineering Survey	June/July 2024
36	BCV303	Engineering Geology	June/July 2024
37	BCV304	Water Supply and Waste Water Engineering	June/July 2024
38	BCV306C	Sustainable Design Concept for Building services	June/July 2024
39	BCV401	Analysis of Structures	June/July 2024
40	BCV402	Fluid Mechanics and Hydraulic s	June/July 2024
41	BCV403	Transportation Engineering	June/July 2024
42	BCV405C	Concreting Techniques and Practices	June/July 2024

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18CV32

Third Semester B.E. Degree Examination, June/July 2024 Strength of Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define and explain four elastic constants. (06 Marks)
b. With neat sketch define the salient features of stress strain curve for tensile specimen. (06 Marks)
c. A bar of a rectangular section of $20\text{mm} \times 30\text{mm}$ and a length of 500mm is subjected to axial compressive load of 60 kN . If $E = 102\text{ kN/mm}^2$ and Poisson's ratio = 0.34 , determine the changes in the length and sides of bar. (08 Marks)

OR

- 2 a. Explain temperature stress induced in body and explain the parameters involved in the expression. (10 Marks)
b. A steel rod of 20mm diameter passes centrally through a copper tube of 50mm external diameter and 40mm internal diameter. The tube is closed at each end and the nuts are tightened on the projecting points of rod. If the temperature of the assembly is raised by 50° . Calculate the temperature stresses developed in copper and steel.
Take $E_s = 200\text{ GN/m}^2$, $E_c = 100\text{ GN/m}^2$ and $\alpha_s = 12 \times 10^{-6}$ per $^\circ\text{C}$ and $\alpha_c = 18 \times 10^{-6}$ per $^\circ\text{C}$. (10 Marks)

Module-2

- 3 a. Derive the expressions for normal and tangential stress on any arbitrary plane inclined at an angle θ with the plane containing σ_x . (10 Marks)
b. At a point in a strained material, the stresses acting are shown in Fig.Q3(b). Determine
i) Principal stresses and their plane
ii) Maximum shear stress and its plane
iii) Normal and shear stress on the inclined plane AB.

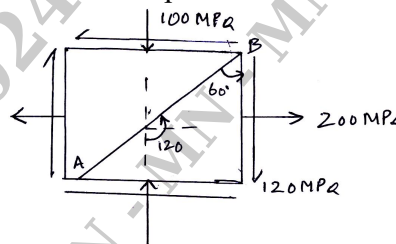


Fig.Q3(b)

(10 Marks)

OR

- 4 a. Determine the expression for hoop stress in thin cylinder with the aid of neat sketch. (06 Marks)
b. A thick cylinder of internal diameter 200mm is subjected to an internal fluid pressure of 40 MPa . If the allowable stress in tension for the material is 120 MPa , find thickness of cylinder. (14 Marks)

Module-3

- 5 a. Derive the relationship between load intensity shear force and bending moment. (05 Marks)
b. Draw SFD and BMD for a cantilever beam of span length ' l ' carrying a point load ' W ' at its free end. (05 Marks)

- c. Draw SFD and BMD for a beam as shown in the Fig.Q5(c).

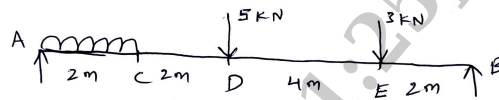


Fig.Q5(c)

(10 Marks)

OR

- 6 a. Draw the shear force and bending moment diagram for beam shown in Fig.Q6(a) marking the salient points. Locate the point of contraflexure and point of maximum bending moment.

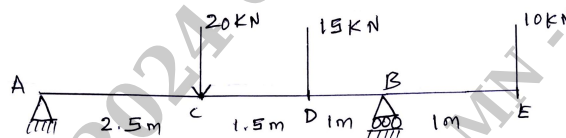


Fig.Q6(a)

(10 Marks)

- b. For the cantilever beam shown in Fig.Q6(b), obtain shear force and bending moment diagrams.

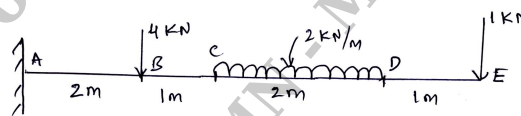


Fig.Q6(b)

(10 Marks)

Module-4

- 7 a. Derive the expression $\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$ for pure bending. (10 Marks)
- b. A rectangular beam 100mm wide and 250mm deep is subjected to a maximum shear force of 50 kN. Determine :
- Average shear stress
 - Maximum shear stress
 - Shear stress at a distance of 25mm above the neutral axes.

(10 Marks)

OR

- 8 a. Derive the torsion equation with usual notations. (10 Marks)
- b. A solid shaft is subjected to a maximum torque of 15 kN-m. Estimate a suitable diameter of the shaft. If the allowable shear stress and the twist are limited to 60 N/mm² and 1° respectively for a length of 20 times the diameter of shaft. Assume $G = 8 \times 10^4$ N/mm². (10 Marks)

Module-5

- 9 a. Derive the expression $EI \cdot \frac{d^2y}{dx^2} = M$, with usual notations. (08 Marks)
- b. Determine the deflection under the loads for beam as shown in the Fig.Q9(b). Take $I = 1.85 \times 10^6$ mm⁴ and $E = 2 \times 10^5$ N/mm².

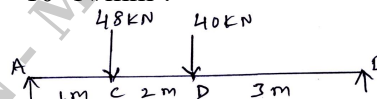


Fig.Q9(b)

(12 Marks)

OR

- 10 a. Derive an expression for buckling load in a column subjected to an axial compressive load, when both ends of the column are hinged. (08 Marks)
- b. Compare the crippling loads obtained by Euler's and Rankine's theory for an axially loaded hollow cylindrical cast column with external diameter and 20mm thick. The column is long and hinged at both ends. Given $\sigma_{\text{failure}} = 550$ MPa, Rankine's constant "a" = 1/1600 and $E = 80$ GPa. (12 Marks)

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18CV35

Third Semester B.E. Degree Examination, June/July 2024 Basic Surveying

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Surveying and mention its main objective. (05 Marks)
- b. Classify surveying according to instruments used. (05 Marks)
- c. The area of the plan of an old survey plotted to a scale of 10 metres to 1 cm measures now as 100.2 cm^2 as found by a planimeter. The plan is found to have shrunk so that a line originally 10 cm long now measures 9.7cm only. There was also a note on the plan that the 20 m chain used was 8 cm too short. Find the true area of Survey. (10 Marks)

OR

- 2 a. Define Ranging and explain the methods of ranging. (05 Marks)
- b. Explain the process of measuring the distance directly along a sloping ground to compute the horizontal distance between two points. (05 Marks)
- c. A steel tape 20m long standardized at 55°F with a pull of 10kg was used for measuring a base line. Find the correction per tape length, if the temperature at the time of measurement was 80°F and the pull exerted was 16kg. Weight of one cubic cm of steel : 7.86 g ; weight of tape : 0.8 kg ; $E = 2.109 \times 10^6 \text{ kg/cm}^2$. Coefficient of expansion of tape per $^\circ\text{F} = 6.2 \times 10^{-6}$. (10 Marks)

Module-2

- 3 a. Define (i) Whole Circle bearing (ii) Reduced Bearing. (05 Marks)
- b. Explain (i) Fore bearing (ii) Back Bearing of a line. (05 Marks)
- c. The following bearings were observed with a compass. Determine the Interior angles.

LINE	F.B.
AB	$60^\circ 30'$
BC	$122^\circ 00'$
CD	$46^\circ 00'$
DE	$205^\circ 30'$
EA	$300^\circ 00'$

(10 Marks)

OR

- 4 a. Define dependent and independent coordinates as applied to traverse survey. (05 Marks)
- b. Explain with sketches latitude and departure of a survey line. (05 Marks)
- c. Calculate latitudes, departures and closing error for the following traverse and adjust the traverse by Bowditch's rule:

LINE	LENGTH (m)	W.C.B.
AB	89.31	$45^\circ 10'$
BC	219.76	$72^\circ 05'$
CD	151.18	$161^\circ 52'$
DE	159.10	$228^\circ 43'$
EA	232.26	$300^\circ 00'$

(10 Marks)

Module-3

- 5 a. Define (i) Back sight (ii) Foresight (iii) Intermediate sight (iv) Change point (v) Station (05 Marks)
- b. Explain temporary adjustments of Dumpy level. (05 Marks)
- c. The following consecutive readings were taken with a level using a 4m levelling staff on a continuously sloping ground at a common interval of 30m:
 0.780, 1.535, 1.955, 2.430, 2.985, 3.480,
 1.155, 1.960, 2.365, 3.640, 0.935, 1.045,
 1.630 and 2.545
 The reduced level of first station A was 180.750. Rule out a page of level field book and enter the above readings. Calculate the reduced levels of all stations by Height of Instrument method. Apply usual check. Also calculate the gradient of the line joining First and Last station. (10 Marks)

OR

- 6 a. Define (i) Check levelling (ii) Reciprocal levelling. (05 Marks)
- b. Compare Height of Instrument method and rise and fall methods of reduction of levels. (05 Marks)
- c. Fill up the missing readings and apply check for a level book page.

Stn	BS	IS	FS	Rise	Fall	R.L.	Remarks
1	3.125					?	BM
2	?		?	1.325		125.505	T.P
3		2.320			0.055	?	
4		?		?		125.850	
5	?		2.655		?	?	T.P
6	1.620		3.205		2.165	?	T.P
7		3.625			?	?	
8			?			123.090	T.B.M

(10 Marks)

Module-4

- 7 a. List any four advantages and disadvantages of plane table survey. (05 Marks)
- b. Explain with sketches the use of (i) Alidade (ii) Plumbing fork, in plane table survey. (05 Marks)
- c. Explain with sketches radiation method of plane tabling. (10 Marks)

OR

- 8 a. Define back orientation in plane table survey and mention the methods of back orientation. (05 Marks)
- b. Explain the operations (i) Centering (ii) Levelling (iii) Orientation as applied to plane table survey. (05 Marks)
- c. Explain intersection method of plane tabling with sketch. (10 Marks)

Module-5

- 9 a. Define (i) Trapezoidal rule (ii) Simpson's one third rule. (05 Marks)
- b. Explain any five characteristics of contours. (05 Marks)
- c. A railway embankment is 10m wide with side slopes $1\frac{1}{2} : 1$. Assuming the ground to be level in a direction transverse to the central line, calculate the volume contained in a length of 120 metres, the central heights at 20m intervals being in metres are :
 2.2, 3.7, 3.8, 4.0, 3.8, 2.8, 2.5.
 Use (i) Trapezoidal (ii) Simpson's rule. (10 Marks)

OR

- 10 a. Define (i) Contour interval (ii) Horizontal Equivalent (05 Marks)
 b. List any five user of contour maps. (05 Marks)
 c. The following table gives the corrected latitudes and departures (in metres) of the side of a closed traverse ABCD.

Side	LATITUDE		DEPARTURE	
	N	S	E	W
AB	108		4	
BC	15		249	
CD		123	4	
DA	0			257

Assuming the coordinates of A as (100, 100), calculate the area by using coordinate method. (10 Marks)

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18CV43

Fourth Semester B.E. Degree Examination, June/July 2024 Applied Hydraulics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State and explain Buckingham's π theorem. Why this theorem is considered superior over the Rayleigh's method. (06 Marks)
- b. Define the following non-dimensional models:
(i) Reynold's model (ii) Froude's model (04 Marks)
- c. The drag force exerted by a flowing fluid on a solid body depends upon the length of the body L , velocity V , density of fluid ρ , and viscosity μ . Find the expression for the drag force using Buckingham's π theorem. (10 Marks)

OR

- 2 a. Explain the term dimensionally homogenous equation with an example. (04 Marks)
- b. Explain Metacentre and Metacentric height with sketches. (08 Marks)
- c. A solid cylinder of diameter 4.0 m has a height of 4m. Find the metacentric height of cylinder if specific gravity of the material of cylinder = 0.6 and it is floating in water with its axis vertical. State whether the equilibrium is stable or unstable. (08 Marks)

Module-2

- 3 a. Explain the terms: (i) Rapidly varied flow (ii) Gradually varied flow. Give examples with sketches. (10 Marks)
- b. Explain most economical channel section. What is the general condition to be satisfied in such a case? Find the slope of the bed of a rectangular channel of width 5m when depth of water is 2m and rate of flow is given as 20 m³/sec. Take Chezy's constant $c = 50$. (10 Marks)

OR

- 4 a. What is specific energy curve? Explain with a sketch. Derive expressions for critical depth and critical velocity in a rectangular channel. (10 Marks)
- b. The discharge of water through a rectangular channel of width 8m is 15 m³/sec. When depth of flow of water is 1.2 m. Calculate:
(i) Specific energy of the flowing water
(ii) Critical depth and critical velocity IV value of minimum specific energy. (10 Marks)

Module-3

- 5 a. Define the term Afflux and Backwater curve. Prove that the length of backwater curve is given by $L = \frac{E_2 - E_1}{i_b - i_e}$. (10 Marks)
- b. A hydraulic jump forms at the downstream end of spillway carrying 17.93 m³/sec discharge. If the depth before jump is 0.8 m, determine the depth after the jump and energy loss. (10 Marks)

OR

- 6 a. Define hydraulic jump. Derive an expression for depth of hydraulic jump. (10 Marks)
 b. Find the slope of the free water surface in a rectangular channel of width 20 m having depth of flow 5m. The discharge through the channel is $50 \text{ m}^3/\text{sec}$. The bed of the channel is having slope of 1 in 4000. Take the value of Chezy's constant $c = 60$. (10 Marks)

Module-4

- 7 a. State Impulse-Momentum equation. Derive an expression for the force exerted by a jet of water on a fixed vertical plate in the direction of the jet. (10 Marks)
 b. A jet of water having a velocity of 20 m/sec strikes a curved vane which is moving with a velocity of 10 m/sec. The jet strikes makes an angle of 20° . With the direction of motion of vane at inlet and leaves at an angle of 130° to the direction of motion of vane an outlet. Calculate: (i) Vane angles, so that the water enters and leaves the vane without shock
 (ii) Work done per second per unit weight if water striking the vane per second. (10 Marks)

OR

- 8 a. Sketch and explain general layout of a hydro-electric plant. (06 Marks)
 b. Explain in detail, how hydraulic turbines are classified. (04 Marks)
 c. A pelton wheel is to be designed for the following specifications:
 Shaft power = 735 KW, head = 200 m, speed = 600 rpm, overall efficiency = 0.75, the jet diameter not to exceed the length of wheel diameter, $C_v = 0.958$, speed ratio = 0.5. Determine:
 (i) Wheel diameter (ii) Number of jets required
 (iii) Diameter of jet (iv) Number of buckets (10 Marks)

Module-5

- 9 a. Sketch and explain Francis Turbine. (10 Marks)
 b. A Kaplan turbine working under a head of 20 m develops 11772 K.Watt shaft power. The outer diameter of runner is 3.5 m and hub diameter 1.75 m. The guide blade angle at the extreme edge of runner is 35° . The hydraulic and overall efficiency of the turbine are 88% and 84%. If the velocity of whirl is zero at outlet, determine:
 (i) Runner vane angles at inlet and outlet at the extreme edge of the runner
 (ii) Speed of turbine (10 Marks)

OR

- 10 a. With a neat sketch, explain the working of a single stage centrifugal pump. (10 Marks)
 b. A centrifugal pump having overall efficiency 75% discharging water at the rate of 30 lit/sec through a pipe of 15 cm diameter and 125 m long. Calculate the power required to drive the pump if the water is lifted to a height of 25 m. Take $f = 0.01$. (10 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024

Analysis of Indeterminate Structures

Time: 3 hrs.

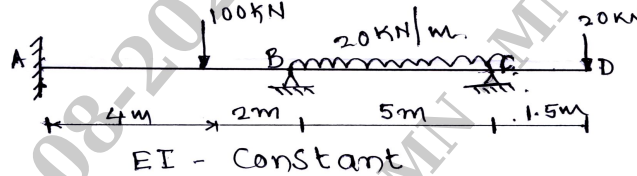
Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 Analyze continuous beam ABCD by slope deflection method. Construct SFD and BMD.

Fig.Q1

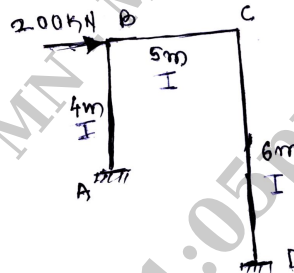


(20 Marks)

OR

- 2 Analyze the frame shown in Fig.Q2 by slope deflection method and draw BMD.

Fig.Q2



(20 Marks)

Module-2

- 3 Analyze the beam shown in Fig.Q.3 by moment distribution method. Draw BMD EI is constant. (20 Marks)

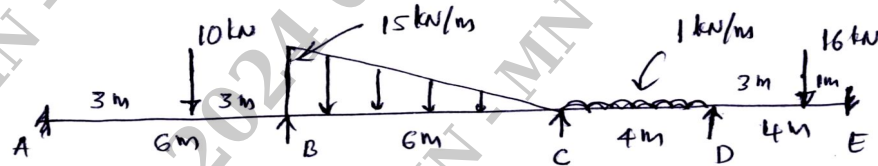
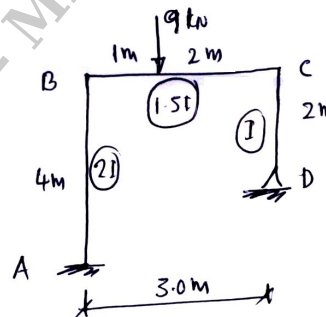


Fig.Q.3.

OR

- 4 Analyze the portal frame by moment-distribution method draw BMD. (20 Marks)

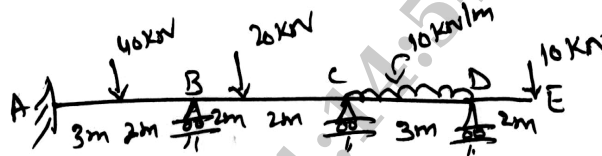
Fig. Q4.



Module-3

- 5 Analyse the continuous beam shown in Fig Q5 by moment distribution method and draw BMD.

Fig. Q5.

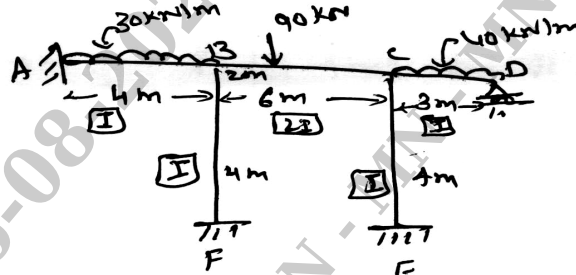


(20 Marks)

OR

- 6 Analyse the frame shown in Fig Q6 by moment distribution method and draw BMD and SFD.

Fig. Q6.

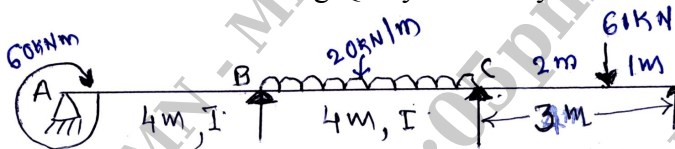


(20 Marks)

Module-4

- 7 Analyze the continuous beam shown in Fig.Q7 by flexibility method. Draw BMD.

Fig.Q7

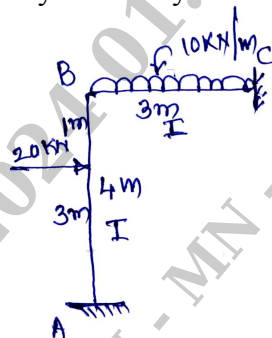


OR

- 8 Analyze frame shown in Fig.Q8 by flexibility matrix approach. Draw BMD.

(20 Marks)

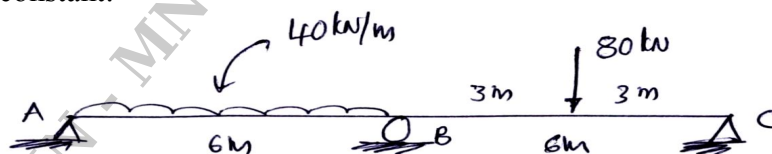
Fig.Q8

**Module-5**

- 9 Analyze the continuous beam by stiffness matrix method (system approach) shown in Fig.Q.9. Draw BMD EI is constant.

(20 Marks)

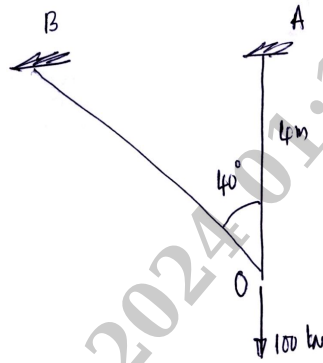
Fig. Q9.



OR

- 10 Find the forces in the members of a joint 'O' shown in Fig.Q.10 by stiffness matrix method. (system approach). (20 Marks)

Fig. Q10.



$$A = 200 \text{ mm}^2 \text{ for } OA \text{ \& } OB$$

$$E = 200 \text{ kN/mm}^2$$

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18CV54

Fifth Semester B.E. Degree Examination, June/July 2024 Basic Geo Technical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following from the phase diagram :
i) Void ratio ii) porosity ii) degree of saturation iv) percent air voids v) air contact. (05 Marks)
- b. Prove the following relationship from the phase diagram :
$$r_b = \left(\frac{G_s + S.e}{1 + e} \right) r_w$$
 (07 Marks)
- c. The soil has porosity of 40% and specific gravity of 2.7. Then calculate void ratio, dry unit weight (r_d) and unit weight for 50% saturation and 100% saturation. (08 Marks)

OR

- 2 a. Classify the soil is well graded or poorly graded from the sieve analysis. (07 Marks)
- b. Define the following indices :
i) Plasticity index
ii) Liquidity index
iii) Consistency index
iv) Toughness index
v) Density index. (05 Marks)
- c. Write a note on ISCS for classification of soil and also explain the use of plasticity chart with an example. (08 Marks)

Module-2

- 3 a. Describe single grained, honey combed flocculent and dispersed structures. (06 Marks)
- b. Write a note on diffuse double layer. (06 Marks)
- c. Explain 3 different clay minerals with neat sketches. (08 Marks)

OR

- 4 a. Write a note on factors affecting compaction. (05 Marks)
- b. How do you calculate placement water context by using protocol's needle method? (07 Marks)
- c. Plot the compaction curve and determine OMC and maximum dry density, also determine the degree of saturator and percentage of air voids at maximum dry density. The SPCT values are as given below :

W.C%	5	10	14	20	25
Bulk density (kN/m ³)	17.6	19.8	21.0	21.7	21.5

(08 Marks)

Module-3

- 5 a. Write a note on confined pumping flow test for calculation of coefficient of permeability. (06 Marks)
- b. How do you calculate the coefficient of permeability for horizontal flow in the case of stratified soils? (06 Marks)
- c. A constant head permeability test was run as a sand sample of 16cm in length and 60cm² in c/s area. Porosity was $n_1 = 40\%$ under a constant head of 30cm, the discharge was found to be 45cm³ in 18sec. Calculate the coefficient permeability. Also determine the discharge velocity and seepage velocity during the test. Estimate the permeability of the sand for a porosity of $n_2 = 35\%$. (08 Marks)

OR

- 6 a. Explain the characteristics of flow nets with a neat sketch. (06 Marks)
- b. How do you locate the phreatic line in a day with filth by using a Casagrande's method? (10 Marks)
- c. Write a note on quick sand phenomena. (04 Marks)

Module-4

- 7 a. Write a note on Mohr-coulomb failure criteria. (08 Marks)
- b. List out the factors affecting the shear strength of soils. (06 Marks)
- c. Write a note on sensitivity and thixotropy. (06 Marks)

OR

- 8 a. What are the advantages of triaxial test over direct shear test? (06 Marks)
- b. Write a note on vane –shear test. (06 Marks)
- c. A series of direct shear tests was conducted as a soil, each test was carried out till the sample failed. The results are as following. Determine the cohesion intercept and the angle of shearing resistance.

Sample no. 1	Normal stress (KPa)	Shear stress (KPa)
1	15	18
2	20	25
3	45	32

(08 Marks)

Module-5

- 9 a. What are the assumptions and limitations of Terzaghi's 1 – D consolidation theory? (08 Marks)
- b. Define the following : i) C_c ii) C_v iii) a_v iv) m_v . (06 Marks)
- c. Explain the Casagrande's method for data mining the pre-consolidation pressure. (06 Marks)

OR

- 10 a. Define normally consolidation over consolidate and under consolidated soils. (06 Marks)
- b. Write a note on square root of time fitting method for calculation of coefficient of consolidation. (08 Marks)
- c. In a consolidation test on a soil, the void ratio of the sample decreases from 1.242 to 1.12 when the pressure is increased from 20 to 40 tons/m². Calculate the coefficient of consolidation. Given that the coefficient of permeability of the soil during this pressure increases, $K = 8.5 \times 10^{-3}$ cm/sec. (06 Marks)

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18CV55

Fifth Semester B.E. Degree Examination, June/July 2024 Municipal Waste Water Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain methods of sewage disposal and systems of disposal. (10 Marks)
- b. Define Dry and Wet weather flow and discuss the factors affecting dry and wet weather flow. (10 Marks)

OR

- 2 a. Explain man hole with sketch. (10 Marks)
- b. With the neat sketch, explain the basic principles of house drainage works. (10 Marks)

Module-2

- 3 a. Explain Self cleaning, Non – scouring and regime velocity in sewage flow. (10 Marks)
- b. The main sewer was designed for an area of 50km². Density of population of town is 200 persons/hectare. The average flow is 250 lpcd. The peak discharge is 1.5 times more than average flow. Rainfall equivalent of 8mm in 24 hours, all of which are runoff. Determine i) The capacity of sewer ii) Min. velocity and gradient required to transport sewage containing coarse sand of 1mm dia. through a sewer of 35cm dia, sp. Gr of particle is 2.65 and value of K = 0.06, f = 0.03. (10 Marks)

OR

- 4 a. Explain the municipal waste water treatment process with flow diagram. (10 Marks)
- b. What are the physical, chemical and biological characteristics of sewage? (10 Marks)

Module-3

- 5 a. Explain the importance of screens and types of screens in the sewage treatment process. (10 Marks)
- b. Explain the i) Sewage forming ii) Grit chamber. (10 Marks)

OR

- 6 a. Explain Oxygen sag curve and zones of purification. (10 Marks)
- b. Sewage flow of 100 cumec from a city is discharged in a perennial river which is fully saturated with oxygen and flows at a min rate of 1250 cumec with a min velocity of 0.15m/sec in the 5 day BOD of the sewage is 260mg/l. Find out where the critical DO will occur in the river. Assume coefficient of purification of river as 4.0, coefficient of DO as 0.11, the ultimate BOD as 125% of the 5 day BOD of the mixture of sewage and river water, temp of 20°C for which DO is 9.17mg/l and DO of effluent as zero. (10 Marks)

Module-4

- 7 a. Explain Activated Sludge processes. (10 Marks)
- b. Design a set of two circular sedimentation tanks to treat 5 Mltrs of sewage per day. Assume detention period of 2 hours. (10 Marks)

OR

- 8 a. Explain sludge digestion areas with a neat sketch. (10 Marks)
b. Design a low rate filter to treat 6.00 MLD of sewage of BOD of 210 mg/ℓ. The final effluent should be 30mg/ℓ and organic loading rate is 320g/m³/d. Assume 30% of BOD load removed in primary sedimentation. (10 Marks)

Module-5

- 9 a. Explain how cost treatment process of sewage. (10 Marks)
b. Explain principle and design of septic tank. (10 Marks)

OR

- 10 a. Explain Eco toilets and its process. (10 Marks)
b. Explain Soak pit with neat sketch. (10 Marks)

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18CV56

Fifth Semester B.E. Degree Examination, June/July 2024 Highway Engineering

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. HRB/PRA soil classification table is allowed.**

Module-1

- 1 a. Explain the role of transportation in social and economic development of the country. (06 Marks)
- b. Explain the recommendations of Jayaker-Committee. List the implementation of Jayakar – Committee. (06 Marks)
- c. The area of a certain district in India is 82,000km² and here are 85 towns as per 1981 census. Determine the lengths of different categories of roads to be provided in third twenty year road development plan. [Density and road 82km/100km² area]. (08 Marks)

OR

- 2 a. Draw a neat sketch :
 - i) Radial or star and circular (06 Marks)
 - ii) Radial or star and grid pattern. (08 Marks)
- b. Explain with sketches the various factors controlling the alignment of roads. (08 Marks)
- c. There are five alternate proposals of road plans for a backward district. The details are given below. Utility units of 0.5, 1, 2, 4 and 8 for the live population ranges and 1 and 5 per 1000A of agricultural and industrial products served.

Proposal	Road length km	Number of town and villages served population					Productivity	In 10001
		< 2000	2001 – 5000	5001 – 10,000	10001 – 20000	>20,000		
P	500	100	150	40	20	3	150	20
Q	600	200	250	68	28	3	220	25
R	700	270	350	82	36	4	300	35
S	800	280	410	91	41	4	400	42
T	900	290	430	96	44	4	430	45

(06 Marks)

Module-2

- 3 a. Define camber. What are the objects of camber? Discuss the factors on which the amount of camber to be provided depends. Specify the recommended ranges of camber for different types of pavement surfaces. (10 Marks)
- b. The speed of overtaking and overtaken vehicles are 70kmph and 40kmph respectively on a two – way traffic road. If the acceleration of overtaking vehicle is 0.99m/sec².
 - i) Calculate safe overtaking distance
 - ii) Mention the minimum length of overtaking zone
 - iii) Draw a neat sketch of the overtaking zone and show the positions of the sign posts. (10 Marks)

OR

- 4 a. Explain PIEV theory. (04 Marks)
 b. Derive an equation for finding the super elevation required if the design co-efficient of lateral friction is 'f'. (10 Marks)
 c. A vertical summit curve is formed at the intersection of two gradients, +3 and – 5.0%. Design the length of summit curve to provide a stopping sight distance for a design speed of 80kmph. Assume other data. (06 Marks)

Module-3

- 5 a. List and explain desirable properties of an aggregates to be used in pavement construction. (06 Marks)
 b. List and explain desirable properties of subgrade soil. (06 Marks)
 c. The properties of subgrade soil are given below passing 0.074mm sieve = 55%, LL = 50%, PL = 41%.
 i) Classify the soil by revised PRA/HRB system.
 ii) Discuss the suitability of the soil as a subgrade material. (08 Marks)

OR

- 6 a. Differentiate between Bitumen and Tar. (06 Marks)
 b. With a neat sketch explain the ESWL in pavement design. (06 Marks)
 c. A load penetration values DBR tests conducted on a specimen of a soil sample are given below. Determine the CBR value of soil, if 100 division of load represents 190kg and in calibration chart of providing ring.

Penetration of plunger in mm	Load Dial Readings (Divisions)
0	0
0.5	0.5
1.0	1.5
1.5	2.5
2.0	6.0
2.5	13
3.0	20
4.0	30
5.0	38
7.5	50
10.0	58
12.5	63

(08 Marks)

Module-4

- 7 a. Explain Rothfutch's method used in design of soil aggregate mix. (10 Marks)
 b. Explain the construction steps for cement concrete pavement slab. (10 Marks)

OR

- 8 a. Explain the construction steps for water bound Macadam roads. (10 Marks)
 b. Write short notes on the following :
 i) Bituminous Macadam
 ii) Prime coat
 iii) Tack coat
 iv) Bituminous concrete. (10 Marks)

Module-5

- 9 a. What are the requirements of highway drainage system? (04 Marks)
b. Explain with sketch how the subsurface drainage system is provided to lower the water table and control seepage flow. (10 Marks)
c. Briefly explain the types of cross-drainage structures. (06 Marks)

OR

- 10 a. Explain the various highway user benefits of highway improvements. (06 Marks)
b. Describe the various methods of economic analysis of a highway. (04 Marks)
c. Compare the annual costs of two types of pavement structures :
i) WBM with thin bituminous surface at total cost of Rs. 2.2 lakhs per km, life of 5 years, interest at 10%, salvage value of Rs. 0.9 lakhs after 5 years. Annual average maintenance cost of Rs. 0.35 lakhs per km and
ii) Bituminous macadam base and bituminous concrete surface, total cost of Rs. 4.2 lakhs life of 15 years interest at 8%, salvage value of 2 lakhs at the end of 15 years, annual average maintenance cost Rs. 0.25 lakhs per km. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Design of Steel Structural Elements

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS : 800 – 2007, SP – 6 – I or steel table is permitted.

Module-1

- 1 a. What are the advantages and disadvantages of steel structures? (10 Marks)
- b. Identify plastic hinge 't' is 0-414 from simple support of propped cantilever beam supporting a UDL of W kN/m over the entire span. (10 Marks)

OR

- 2 a. What are rolled steel sections? Mention any six shapes used as a structural elements with sketches. (08 Marks)
- b. Analyze the continuous beam ABC shown in Fig Q2(b) and calculate maximum plastic moment. Consider load factor as 1.5.

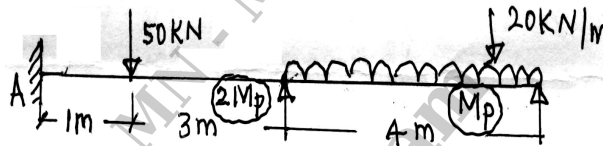


Fig Q2(b)

(12 Marks)

Module-2

- 3 a. Define : i) Pitch of bolt ii) Gauge distance iii) Edge distance iv) Staggered distance. (08 Marks)
- b. Determine the bolt value used in a lap joint connecting two plates of thickness 10mm and 8mm. Use M16 bolts with property class 4-6 and strength of plate 410N/mm^2 . Assume $e = 30\text{mm}$, $p = 50\text{mm}$ and partially threaded bolts are used. (12 Marks)

OR

- 4 a. What are the advantages and disadvantages of welded connections? (12 Marks)
- b. Two plates of 16mm and 14mm are to be joined by a butt weld and joint is subjected to factored tensile force of 430kN. Due to some reasons effective length of weld that to be provided is 175mm. Check the safety of the joint for the following cases.
i) Single V – Butt weld
ii) Double V – Butt weld (08 Marks)

Module-3

- 5 a. Explain Laced and Battened columns with sketches. (08 Marks)
- b. Design a column section using single rolled steel beam along with cover plates to carry of factored load of 2000kN. Column both ends are fixed with following data :
Section : ISHB150@ 30.6Kg/m
Cover plates : 250mm $\times$ 20mm
Take $\ell = 6\text{m}$. (12 Marks)

OR

- 6 Design a compression member using double channel section (2ISLC 300@33-1Kg/m) face to face to carry factored load of 1600kN. The length of the column is 5m with one end fixed and one end hinged. Assume M18 bolts and $f_{cd} = 200\text{N/mm}^2$. Also design single facing system. (20 Marks)

Module-4

- 7 a. Briefly explain Gussetted base with sketches. (06 Marks)
 b. Determine the tensile strength of a tie member ISA $100 \times 75 \times 8\text{mm}$ connected to Gusset plate using 4 bolts in a row of 16mm diameter at a pitch of 60mm and edge distance of 40mm. Take $f_y = 250\text{N/mm}^2$, $f_u = 410\text{N/mm}^2$ and $C = 60$. (14 Marks)

OR

- 8 a. What is lug angle? Explain briefly with sketch. (06 Marks)
 b. Design a slab base for a column ISHB 300@58.8Kg/m subjected to ultimate load of 2250kN. The grade of concrete for pedestal is M_{20} and SBC of soil is 180kN/m^2 . Design slab base and concrete base. (14 Marks)

Module-5

- 9 Design a beam of effective span 6m subjected to UDL of 10kN/m along with 100kN point load. The beam is laterally supported. Thickness of wall is 230mm. Assume self weight of beam as 1kN/m . (20 Marks)

OR

- 10 Simply supported beam [ISMB350@52.4Kg/m](#) is used over a span of 5m. The beam carries an UDL live load of 20kN/m and dead load is 15kN/m . The beam is laterally supported throughout check the safety of the beam. (20 Marks)

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18CV62

Sixth Semester B.E. Degree Examination, June/July 2024 Applied Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS : 6403 is permitted.

Module-1

- 1 a. State the objectives of soil exploration programme. (06 Marks)
b. With a neat sketch, explain the seismic refraction method. (06 Marks)
c. Estimate ground water table, with the following data :
i) Depth upto which water bailed out = 15.0m
ii) Water rise in first day = 0.8m
iii) Water rise in second day = 0.70m
iv) Water rise in third day = 0.60m. (08 Marks)

OR

- 2 a. With a neat sketch, explain auger boring. (06 Marks)
b. With a neat sketch of a sampling tube, define :
i) area ratio ii) recovery ratio iii) inside clearance iv) outside clearance. (06 Marks)
c. List the methods used for ground water control during excavation. Explain any one of these dewatering systems. (08 Marks)

Module-2

- 3 a. Explain construction and uses of Newmark's chart. (06 Marks)
b. A concentrated load of 20kN acts on the surface of a soil mass. Using Boussinesq analysis find vertical stress at points.
i) 3m below axis of loading
ii) At radial distance of 2m from axis of loading but at same depth of 3m. (06 Marks)
c. Explain different types of statements of footings. (08 Marks)

OR

- 4 a. List the assumptions made in Boussinesq and Westergaard theories. (06 Marks)
b. Explain:
i) Pressure bulb
ii) Pressure distribution on horizontal plane
iii) Pressure distribution on vertical plane. (06 Marks)
c. A saturated clay 8m thick underlies a proposed new building. The existing overburden pressure at the centre of clay layer is 250kN/m^2 and due to a new building increases the pressure by 120kN/m^2 . The liquid limit of the soil is 74%. Water content of soil is 40% specific gravity is 2.67. Estimate consolidation settlement. (08 Marks)

Module-3

- 5 a. Explain Culmann's graphical method of finding out the active earth pressure. (06 Marks)
b. What are the causes of failure of slopes? Explain various types of slope failure. (06 Marks)
c. A retaining wall, 6m high retains dry sand with an angle of friction of 30° and unit weight of 16.2kN/m^3 . Determine the earth pressure of rest. If the water table rises to the top of the wall, determine the increase in the thrust on the wall. Assume submerged unit weight of sand as 10kN/m^3 . (08 Marks)

OR

- 6 a. Explain method of slice to determine the factor of safety against failure of finite slope. (06 Marks)
- b. Explain Rankine's theory for calculating active earth pressure in cohesionless soils for no surcharge. (06 Marks)
- c. A slope 1 in 2 with a height of 8m has the following soil properties $C = 28\text{kN/m}^2$, $\phi = 10^\circ$, $\gamma = 18\text{kN/m}^3$. Calculate :
- Factor of safety with respect to cohesion
 - Critical height of slope
- Table of Taylor's stability number to be given for question 6(c).

$\phi \rightarrow$ i ↓	0°	5°	10°	15°
45°	0.170	0.136	0.108	0.083
30°	0.156	0.110	0.075	0.046
15°	0.145	0.068	0.023	—

(08 Marks)

Module-4

- 7 a. Define :
- Ultimate bearing capacity
 - Net ultimate bearing capacity
 - Safe bearing capacity.
- b. Explain with a neat sketch, general shear failure. (06 Marks)
- c. A rectangular footing has a size of $1.8\text{m} \times 3.0\text{m}$ has to transmit the load of a column at a depth of 1.5m. Calculate the safe load which the footing can carry at a factor of safety of 3 against shear failure. Use is code method. The properties of soil are : $n = 40\%$, $G = 2.67$, $w = 15\%$, $c = 8\text{kN/m}^2$, $\phi = 32.5^\circ$. (08 Marks)

OR

- 8 a. What are the assumptions made in Terzaghi's analysis to find bearing capacity of soils. (04 Marks)
- b. Explain plate load test with neat sketch to determine bearing capacity of soil. (08 Marks)
- c. A strip footing 2m wide carries a load intensity of 400kN/m^2 at a depth of 1.2m in sand. The saturated unit weight of sand is 19.5kN/m^3 and unit weight above water table is 16.8kN/m^3 . The shear strength parameters are $c = 0$ and $\phi = 35^\circ$. Determine the factor of safety with respect to failure for the following locations of water table : i) water table at 2.5m below ground level ii) Water table is 0.5m below ground level. Take $N_q = 41.4$ and $N_\gamma = 42.4$. (08 Marks)

Module-5

- 9 a. Explain load carrying capacity of pile by i) Dynamic formulae ii) Static formulae. (10 Marks)
- b. A group of 9 piles, 10m long is used as a foundation for a bridge pier. The piles used are 30cm diameter with centre to centre spacing of 0.9m. The sub soil consists of clay with unconfined compressive strength of 15t/m^2 . Determine the efficiency neglecting the bearing action. Given adhesion factor = 0.9. (10 Marks)

OR

- 10 a. With a neat sketch, explain the classification of pile foundation. (08 Marks)
- b. Explain Negative skin friction. (08 Marks)
- c. A wooden pile is being driven with a drop hammer weighing 20kN and having a free fall of 1.0m. The penetration in the last blow is 5mm. Determine the load carrying capacity of the pile according to the engineering News formula. (04 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024

Hydrology and Irrigation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with neat sketch, qualitative representation of hydrological cycle. (10 Marks)
- b. List and explain different types of precipitation. (10 Marks)

OR

- 2 a. Explain with neat sketch, syphon type of raingauge. (10 Marks)
- b. Average annual rainfall of four existing raingauge stations in a basin are 105mm, 79mm, 70mm and 66mm. If the average depth of rainfall over the basin is limited with in 10% error. Determine the additional number of gauges required. (10 Marks)

Module-2

- 3 a. Explain how evaporation amount is measured using IS class-A pan. List the factors affecting it. (10 Marks)
- b. The total observed runoff volume during a 6hr-storm with a uniform intensity of 1.5cm/hr is $21.6 \times 10^6 \text{m}^3$. If the area of the basin is 350km^2 , find the average infiltration rate for the basin. (10 Marks)

OR

- 4 a. List the factors affecting evapo-transpiration. Write Blaney-criddle equation used to estimate evapo-trnspiration. (10 Marks)
- b. With a neat sketch, describe the method of determining infiltration capacity using a double ring infiltrometer. (10 Marks)

Module-3

- 5 a. What is run off? Explain the factors affecting run off. (10 Marks)
- b. The ordinates of a storm hydrograph due to 6h isolated storm is given. Obtain the ordinates of 6h unit hydrograph for the catchment, if its area is 423km^2 .

Time (hr)	0	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90
Discharge (m^3/s)	10	32	88	116	102	85	71	59	47	39	32	26	22	18	15	10

(10 Marks)

OR

- 6 a. Explain with a neat sketch, base flow separation methods. (10 Marks)
- b. Define unit hydrograph. Mention the assumptions and its limitations of unit hydrograph. (10 Marks)

Module-4

- 7 a. Briefly explain the benefits and ill effects of irrigation. (10 Marks)
- b. With a neat sketch, explain Bandhara irrigation. List its advantages and disadvantages. (10 Marks)

OR

- 8 a. Explain the factors affecting duty of water. (10 Marks)
- b. Table gives the necessary data about the crop, their duty and area under each crop, commanded by a canal taking off from a storage tank. Taking time factor for the canal 13/20, calculate the discharge required at the head to the canal. If the capacity factor is 0.8, determine the design discharge.

Crop	Base period (days)	Area (ha)	Duty (ha/cumec)
Sugar cane	320	850	580
Overlap for sugar cane in summer	90	120	580
Wheat (Rabi)	120	600	1600
Bajri (Monsion)	120	500	2000
Veg (Hot weather)	120	360	800

(10 Marks)

Module-5

- 9 a. What is canal? List its types and explain the classifications based on capacity. (10 Marks)
- b. Design the canal for the discharge of 30 cumec with silt factor 1.0. Side slope – 0.5H:1V. (10 Marks)

OR

- 10 a. Explain the investigations of reservoir planning. List the points to be considered for selection of site for a reservoir. (10 Marks)
- b. Explain different storage zones of reservoir with neat sketch. (10 Marks)

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18CV643

Sixth Semester B.E. Degree Examination, June/July 2024

Alternate Building Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the environmental issues related to building materials. (10 Marks)
- b. With neat sketch, explain rain water harvesting enlisting its various advantages. (10 Marks)

OR

- 2 a. Write a note on Green building ratings. (06 Marks)
- b. Discuss buildings in different climatic regions. (06 Marks)
- c. Write short notes on:
 - i) Roof Pond System
 - ii) Direct gain passive solar system. (08 Marks)

Module-2

- 3 a. Explain the method of manufacturing the stabilized mud blocks. (10 Marks)
- b. Explain briefly the use of stones and laterite blocks as masonry units. (10 Marks)

OR

- 4 a. List out the factors influencing compressive strength of masonry. Explain any two. (10 Marks)
- b. Write short notes on:
 - i) FAL-G blocks
 - ii) Characteristics of a good brick. (10 Marks)

Module-3

- 5 a. What is lime-pozzolana cement? Explain the different ways of blending lime and pozzolana. (10 Marks)
- b. List out matrix materials and reinforcing materials in fiber reinforced cement composites and explain their behavior of FRC in compression. (10 Marks)

OR

- 6 a. Explain typical agro wastes and other biomass resources in construction. (10 Marks)
- b. Discuss the following:
 - i) Application of FRC in construction.
 - ii) Production of Rice husk ash in rice husk burner. (10 Marks)

Module-4

- 7 a. Explain various types of filler slabs along with materials. (10 Marks)
- b. Explain masonry vaults and domes. (10 Marks)

OR

- 8 a. Explain the materials used in production of ferro cement products. (10 Marks)
b. Discuss the following in brief:
i) Cavity wall
ii) Rammed Earth. (10 Marks)

Module-5

- 9 a. Explain the methods involved in production of precast elements. (10 Marks)
b. Discuss in brief cost saving techniques in construction. (10 Marks)

OR

- 10 a. Write a note on global warming and construction industries. (10 Marks)
b. Explain the machine used in making of stabilized mud block. (10 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2024
Design of RCC and Steel Structures

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any ONE full question each from Module-1 and Module-2.
 2. Use of IS – 456, IS – 800, SP – 16, SP(6) and steel tables is permitted.
 3. Missing data, if any, may be suitably assumed and same must be stated clearly.

Module - 1

- 1 Design a cantilever Retaining wall to retain an earthen embankment 5m height above GL. Given Density of backfill = 18kN/m^3 ; Angle of internal friction = 30° : Co-efficient of friction between soil and slab = 0.5 and SBC of soil = 200kN/m^2 . Adopt M_{20} grade concrete and Fe 415 steel. Draw a neat sketch of the designed reinforcement details. (50 Marks)

OR

- 2 Design a rectangular combined footing for supporting two columns $400\text{mm} \times 400\text{mm}$ in size to carry a load of 1000kN each. Centre to centre distance between the columns is 3.5m . The projection of the footing on either side of the column with respect to centre is 1m . SBC = 190kN/m^2 . Adopt M_{20} grade of concrete and Fe 415 steel. Draw a neat sketch of the designed reinforcement detail. (50 Marks)

Module – 2

- 3 The line diagram of a truss with loads and tabulation of factored member forces are shown in the Fig Q3. Design the various members of the roof truss along with their end connections with gusset plates by bolts. Assuming the thickness of supporting walls as 450mm , design the supports consisting the shoe angle bearing plate, base plate by considering support reaction and anchor bolts for an uplift force of 15kN at each support. The bearing pressure of concrete may be taken as 4N/mm^2 . [Refer Fig. Q3 and Table Q3]

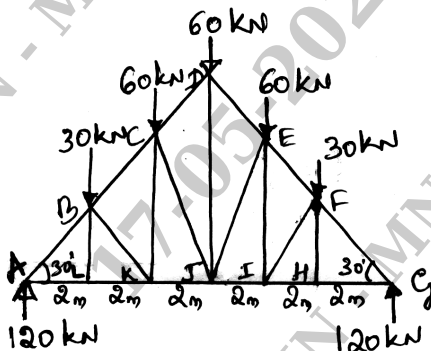


Fig Q3

Members	Length (m)	Force (kN)
AB, GF	2.31	-240
BC, FE	2.31	-210
CD, ED	2.31	-160.04
AL, GH	2	+207.84
LK, HI	2	+207.84
KJ, IJ	2	+181.32
BL, FH	1.154	0
BK, FI	2.31	-30
CK, EI	2.31	+15
CJ, EK	3.05	-66.05
DJ	3.46	-60

Table Q3

(50 Marks)

4. Design a simply supported welded plate girder with thick web without stiffness to carry a UDL of 50kN/m, over a span of 20m. Design the following :

- i) C/s of the girder @ mid span
- ii) Curtailment of flange plates
- iii) Connection details

Draw a neat sketch of designed details.

(50 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2024

Ground Water Hydraulics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the vertical distribution of ground water with neat sketches. (10 Marks)
- b. Explain the perched aquifer with neat sketches. (04 Marks)
- c. Compare to surface water storage of ground water is important why? (06 Marks)

OR

- 2 a. Describe the following with examples:
i) Aquifer ii) Aquifuge iii) Aquiclude iv) Aquitard. (08 Marks)
- b. Describe confined and unconfined aquifers with neat sketches. (12 Marks)

Module-2

- 3 a. What is storage co-efficient explain its characteristics in confined and unconfined aquifers? (12 Marks)
- b. An artesian aquifer 20m thick has a porosity of 20% and bulk modulus of compression 10^8N/m^2 . Estimate the storage coefficient of the aquifer what fraction of this is attributable to the expansibility of water. (08 Marks)

OR

- 4 a. Explain the terms : i) Porosity ii) Specific yield iii) Specific retention
iv) Permeability. (08 Marks)
- b. Explain Darcy's law with derivations. (04 Marks)
- c. In an area of 100 hectares the water table dropped by 4.5m. If the porosity is 30% and the specific retention is 10%. Determine:
i) The specific yield of the aquifer
ii) Change in ground water storage. (08 Marks)

Module-3

- 5 a. Describe the steady radial flow into a well for an unconfined aquifer with neat sketches. (10 Marks)
- b. Pumping at a rate of 1500 lpm from a 30cm diameter test well penetrates into 60m of an unconfined aquifer gives draw-down of 2.0 and 1.10m in observation wells located respectively at 120m and 160m away from it. Calculate:
i) Hydraulic conductivity of the aquifer
ii) Draw down of the pumping well. (10 Marks)

OR

- 6 a. Explain cooper Jacob method of finding aquifer parameter in a confined aquifer of unsteady flow. (10 Marks)
- b. A 40cm well penetrates 40m into an aquifer. Screen is provided for the first 15m. The yield from the well. The yield from the well is 2500 lpm with the draw-down at the well of 4m. If the length of screen is in glazed to 25m. Find the new discharge. (10 Marks)

Module-4

- 7 a. Explain the ground water investigation using electrical resistivity method. (10 Marks)
b. Explain the following:
i) Electrical logging
ii) Radio-active logging. (10 Marks)

OR

- 8 a. Describe the ground water investigation using seismic method. (10 Marks)
b. Describe the following:
i) Induction logging
ii) Sonic logging. (10 Marks)

Module-5

- 9 a. With a neat sketches explain parts and construction process of Dug well with working principles. (10 Marks)
b. Explain the types of tube wells with neat sketches. (10 Marks)

OR

- 10 a. Describe the different methods of ground water recharge. (12 Marks)
b. What is conjunctive use of water? Explain its necessary, advantages and disadvantages. (08 Marks)

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18CV745

Seventh Semester B.E. Degree Examination, June/July 2024 Urban Transport Planning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Urbanisation. What are the various causes of Urbanisation? (10 Marks)
- b. With flow diagram, explain the concept of system approach to transport planning process. (10 Marks)

OR

- 2 a. Describe the details of BRTS. (10 Marks)
- b. Explain the principles of Coordination (10 Marks)

Module-2

- 3 a. Define Zone. What are the factors to be considered for making study area into zones? (10 Marks)
- b. List and explain the types of sampling methods. (10 Marks)

OR

- 4 a. Discuss the details of Home Interview method. (10 Marks)
- b. Define Sampling. Explain i) Cluster sampling ii) Stratified sampling. (10 Marks)

Module-3

- 5 a. With assumptions discuss the details of category analysis. (10 Marks)
- b. The trip rate (y) and the corresponding household sizes (x) from a sample are shown in table below. Compute the trip rate if average household size is 3.25.

	Household size (x)			
	1	2	3	4
Trips per day (y)	1	3	4	5
	3	4	5	8
	3	5	7	8

(10 Marks)

OR

- 6 a. Explain the details of multiple linear regression analysis. Give any two examples. (10 Marks)
- b. Find the trip distribution between the zones using Average growth factor method. Refer the table data. (10 Marks)

O \ D	1	2	3	pi	Pi
1	60	100	200	360	360
2	100	20	300	420	1260
3	200	300	20	520	3120
aj	360	420	520	1300	
Aj	360	1260	3120		4740

Module-4

- 7 a. The total trips produced and attracted to the three zones A, B & C of a survey area in the design year are tabulated as :

Zone	Trips produced	Trips attracted
A	2000	3000
B	3000	4000
C	4000	2000

It is known that the trips between the two zones are inversely proportional to the second power of the travel time between the zones, which is uniformly 20 minutes. If the trip interchange between zones B & C known to be 600, calculate the trip interchange between zones A & B , A & C , B & A and C & B. (10 Marks)

- b. Briefly explain the details of i) Tanner model ii) Intervening opportunity model. (10 Marks)

OR

- 8 a. List and explain the factors affecting on modal split. (10 Marks)
 b. Briefly explain the details of :
 i) Gravity model ii) Desire line diagram. (10 Marks)

Module-5

- 9 a. Discuss the details of diversion curves. (10 Marks)
 b. Explain the general principles of traffic assignment. (10 Marks)

OR

- 10 Briefly discuss the details of :
 a. All or nothing assignment.
 b. Capacity restraint assignment.
 c. Equilibrium assignment.
 d. Land use planning models. (20 Marks)

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Eighth Semester B.E. Degree Examination, June/July 2024 Design of Pre-Stressed Concrete

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Assume and indicate missing data, if any.**

Module-1

- 1 a. Explain why high strength steel and high strength concrete are used in pre-stressed concrete. (10 Marks)
- b. Explain with a neat sketch, "Hoyer's line" system of pre-tensioning. (10 Marks)

OR

- 2 a. Explain pressure line concept. (10 Marks)
- b. A beam of symmetrical I section spanning 12m has a flange width of 300mm and thickness 80mm. The web depth is 80mm. The overall depth of the beam is 800mm. The parabolic cable has an eccentricity of 300mm at the centre and zero eccentricity at the end and it carries an initial pre-stressing force of 150kN. The live load on the beam is 3.5kN/m. Assuming a loss of 20% by strength concept, calculate the stress at the mid span section during the transfer of pre-stress and working load condition. Assume density of concrete as 25kN/m³. (10 Marks)

Module-2

- 3 a. List and explain the various losses in pre-stressed concrete. (06 Marks)
- b. A pre-stressed concrete beam of size 200mm × 300mm is pre-stressed with $A_s = 160\text{mm}^2$ to an initial pre-stress of 1000N/mm² at a constant eccentricity of 50mm. The beam spans 10m. Calculate the percentage pre-stress loss if the beam is post tensioned considering the wires simultaneously tensioned. $E_s = 210\text{ kN/mm}^2$, $E_c = 35\text{ kN/mm}^2$ relation of stress in steel is 5% of initial stress, Shrinkage of concrete is 200×10^{-6} , slip and deformation is 2.0, wobble co-efficient = 0.0015/m. (14 Marks)

OR

- 4 a. Discuss the various factors affecting pre-stressed concrete beams. (08 Marks)
- b. A concrete beam with a cross sectional area of $32 \times 10^3\text{mm}^2$ and radius of gyration of 72mm is pre-stressed by parabolic cable carrying an effective stress of 1000N/mm². The span of the beam is 3m. The cable composed of 6 wires of 7mm diameter has an eccentricity of 650mm at the centre and zero at the supports neglecting all losses, find the central deflection of the beam for the following cases. Assume : $E_c = 38\text{ kN/mm}^2$, Density = 24kN/m³.
 - i) Self weight + Pre-stress
 - ii) Self weight + pre-stress + live load of 2kN/m. (12 Marks)

Module-3

- 5 a. List the different types of flexural failures in a PSC beam. Explain failure of over reinforced sections. (06 Marks)
- b. A post tensioned bridge girder with unbounded tendons is of base section of overall dimensions 1200mm wide by 1800mm deep with wall thickness of 150mm. The high tensile steel has an area of 4000mm² and is located at an effective depth of 1600mm. The effective pre-stress in steel after losses is 1000N/mm² and the effective span of the girder is 24m. If $f_{ck} = 40\text{ N/mm}^2$ and $f_p = 1600\text{ N/mm}^2$. Estimate the ultimate flexural strength of the section. (14 Marks)

OR

- 6 Design a symmetrical I section of span 16m to carry superimposed load of 18kN/m. Assume compressive stress of concrete as 15MPa at transfer and 12MPa at working load. The permissible tensile stress in concrete at both the stages of loading are 1MPa. Assume the loss of pre-stresses 20% and the initial pre-stress in steel shall not exceed 5000MPa. (20 Marks)

Module-4

- 7 a. Explain the different types of shear failure cracks. (06 Marks)
 b. A simply supported beam of span 6m is 120×300 mm is section. It is pre-stressed with parabolic cable which carries an effective pre-stress of 200kN. The cable has a maximum eccentricity of 100mm at mid span section and minimum eccentricity of 50mm at the support section. Determine the principle tension at 20mm above the centriodal fibre in a section. Determine which lies at 0.6m from the left support. The beam carries an all inclusive load of 15kN/m. (14 Marks)

OR

- 8 a. Explain the following i) Diagonal Tension failure ii) Shear compression Failure. (08 Marks)
 b. The support section of PSC beam $150\text{mm} \times 300\text{mm}$ deep is required to support an ultimate shear force of 100kN. Compressive pre-stress at the centriodal axis is 5N/mm^2 . Adopt M40 grade concrete and cover to tension reinforce as 45mm. Design suitable reinforcements at the section using IS : 1343 recommendations. Use 8mm ϕ 2 legged vertical stirrups. Take $f_y = 250\text{N/mm}^2$. (12 Marks)

Module-5

- 9 a. Explain the stress distribution in end block of a post tensioned pre-stressed concrete member with neat sketch. (06 Marks)
 b. The end block of a PSC girder is 200mm by 300mm . The beam is post-tensioned – tensioned by 2 anchorages each of 100mm diameter with their centers located at 75mm from the top and bottom of the beam. The force and design suitable reinforcements according to IS : B43 provision. Also sketch the arrangement of anchorage zone reinforcement use : 10# links yield stress. (14 Marks)

OR

- 10 a. Explain the factors effecting of End block stress distributions. (06 Marks)
 b. The end block of a pre-stressed concrete girder is 200mm wide by 300mm deep. The beam is post tensioned by two Freyssient anchorages each of 100mm diameter with their centers located at 75mm from the top and bottom of beam. The force transmitted by each anchorage being 2000kN. Compute the bursting force and design suitable reinforcements according to Indian standard IS1343 code provisions. Sketch the arrangement of anchorage zone reinforcement. (14 Marks)

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18CV824

Eighth Semester B.E. Degree Examination, June/July 2024 Rehabilitation and Retrofitting

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the following terms:
 - i) Repair
 - ii) Rehabilitation
 - iii) Retrofitting
 - iv) Strengthening

(08 Marks)
- b. What are the physical causes for deterioration of concrete structures? Explain any two physical causes in detail. (12 Marks)

OR

- 2 a. Explain different evaluation methods of structural damage to the structural elements due to earthquake. (15 Marks)
- b. Explain in detail the deterioration of concrete structures due to sulphate attack. (05 Marks)

Module-2

- 3 a. What are the basic items covered in rapid damage assessment of distressed concrete structures? (10 Marks)
- b. Brief the damage assessment procedure with the help of flow chart. (10 Marks)

OR

- 4 a. Define NDT. List the destructive non destructive and semi-destructive testing systems on concrete structures. (10 Marks)
- b. Explain rebound hammer test and ultra pulse velocity test procedure used to assess the quality of concrete with a neat sketch. (10 Marks)

Module-3

- 5 a. Explain the influence of design and construction errors on durability of concrete. (10 Marks)
- b. Explain the effect of cover thickness and cracking on durability of concrete. (10 Marks)

OR

- 6 a. Explain the corrosion mechanism in reinforced concrete structures with the help of sketch. (10 Marks)
- b. Define corrosion inhibitors and explain different types of corrosion inhibitors. (10 Marks)

Module-4

- 7 a. Define maintenance. Explain the importance of maintenance of concrete structures. (08 Marks)
- b. Explain the causes which necessitate the maintenance, effects the service and durability of concrete structures. (12 Marks)

OR

- 8 a. Discuss :
i) Necessity of jacketing
ii) Advantages of jacketing
iii) Types of jacketing (10 Marks)
b. Discuss ERB and NSM techniques. (10 Marks)

Module-5

- 9 a. Explain the following: i) Epoxy injection ii) Repair mortars. (10 Marks)
b. Mention different types of special concretes. Explain polymer concrete and sulphur infiltrated concrete. (10 Marks)

OR

- 10 Write short notes on:
a. Rust eliminators
b. Guniting and shotcrete
c. Shoring and underpinning
d. Fibre reinforced concrete. (20 Marks)

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21CV32

Third Semester B.E. Degree Examination, June/July 2024 Geodetic Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Surveying. Discuss the classification of Surveying. (04 Marks)
- b. Distinguish between :
 - i) True meridian and Magnetic Meridian.
 - ii) Whole circle bearing and Quadrantal bearing.
 - iii) Fore bearing and Back bearing.
 - iv) Dip and Declination. (08 Marks)
- c. The following bearings were observed with a compass. Calculate the interior angles. (08 Marks)

Line	AB	BC	CD	DA
Fore bearing	N 45° 30' E	S 60° 0' E	S 10° 30' N	N 75° 45' W

OR

- 2 a. Define Local Attraction and explain the procedure of Detection and Elimination methods. (04 Marks)
- b. Describe briefly Radiation and Intersection method of plant labeling. (08 Marks)
- c. The following bearings were observed in running a closed traverse.

Line	AB	BC	CD	DE	EA
FB	75° 5'	115° 20'	165° 35'	224° 50'	304° 50'
BB	254° 20'	296° 35'	345° 35'	44° 05'	125° 05'

At what stations do you suspect local attraction? Determine the correct magnetic bearings. If declination was 5° 10' E, what are the true bearings? (08 Marks)

Module-2

- 3 a. Define i) Back sight ii) Fore sight iii) Intermediate sight iv) Change point. (04 Marks)
- b. Explain the procedure with sketch reciprocal levelling. (08 Marks)
- c. Following readings were taken with a level and a 4m levelling staff on a continuously sloping ground
1.680 , 2.470 , 3.550 , 0.680 , 1.200 , 2.050 , 3.800 , 1.200 , 1.600 , 1.850 , 3.600 , 1.800 , 2.500 , 3.500.
Rule out a page of level book and find out RL of each point by Rise and Fall method. Taking the BM RL as 100.00m. Apply usual checks. (08 Marks)

OR

- 4 a. Discuss the methods of determining areas and volume. (05 Marks)
- b. Explain the temporary adjustments and dumpy level. (05 Marks)

- c. Calculate the volume of earthwork for a road having the following data :

Chainage (m)	Depth of cutting (m)	Transverse slope
0	1.50	1 in 8
40	2.50	1 in 4
80	1.80	1 in 10

If formation width = 10m and side slopes 1 : 1.

(10 Marks)

Module-3

- 5 a. Explain the procedure of measuring horizontal angles by Repetition and Reiteration method. (05 Marks)
- b. Briefly explain the Double plane method of finding the RL of an point. (05 Marks)
- c. To determine the elevation of the top of aerial pole, the following observations were made :

Instrument station	Reading on BM	Angle of Elevation	Remarks
A	1.377	11° 53'	RL of BM = 30.150m
B	1.263	8° 5'	

Stations A & B & top of aerial pole are in the same vertical plane. Find the RL of top of aerial pole. If distance between A & B is 30.00m.

(10 Marks)

OR

- 6 a. Explain the following terms :
 i) Transiting ii) Changing face
 iii) Swinging iv) Line of collimation. (04 Marks)
- b. Derive the expression for distance and elevation for stadia tacheometry, when the staff held vertical to the line of sight, for both angle of elevation and angle of depression. (06 Marks)
- c. The following observations were observed with a tacheometer fitted with anallactic lens and multiplier constant being 100.

Inst. Station	Staff Station	Height of axis	Vertical angle	Hair readings
P	BM	1.5	- 6° 12'	0.963, 1.515, 2.067
P	Q	1.5	7° 5'	0.819, 1.341, 1.863
Q	R	1.6	12° 23'	1.860, 2.445, 3.030

RL of BM is 460.500m, staff held vertically. Compute the RL of P, Q & R and the horizontal distances PQ & QR.

(10 Marks)

Module-4

- 7 a. Explain the following along with a neat sketch :
 i) Forward tangent ii) Point of curve
 iii) Deflection angle iv) Long chord. (04 Marks)
- b. What is the relation between degree and radius of a curve? (06 Marks)
- c. Calculate the necessary data for setting out a curve by Rankine's method and prepare a curve table. Chainage of PI = 1192m, Deflection angle = 50° 30', Radius of curve = 300m, Peg interval = 20m, Theodolite least count 20". (10 Marks)

OR

- 8 a. What are the requirements of a transition curve? (04 Marks)
- b. With a neat sketch, explain the elements of a compound curve. (06 Marks)

- c. Two parallel railway lines are to be connected by reverse curve, each section having the same radius. If the lines are 12m apart and the maximum distance between tangent points measured parallel to the straights is 48m, find the maximum allowable radius. If however, both the radius are to be different, calculate the radius of the second branch if that of the first branch is 60m. Also calculate the length of both the branches. (10 Marks)

Module-5

- 9 a. Derive an expression for relief displacement on vertical photograph. (08 Marks)
b. What are the advantages of LiDAR? (04 Marks)
c. Explain the components of GIS, with neat sketch. (08 Marks)

OR

- 10 a. What is an EDM? Explain the principle used by EDM for distance measurement. (08 Marks)
b. Explain in brief any two applications in Civil engineering of
i) Remote sensing ii) GPS. (04 Marks)
c. A vertical photograph was taken at an altitude of 1200m above MSL. Determine the scale of the photograph for terrain lying at elevations of 80m and 300m, if the focal length of the camera is 15cm. (08 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Strength of Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1.
 - a. With a neat sketch, define salient features of stress-strain curve for a tensile specimen for mild steel. (06 Marks)
 - b. Derive the expression for elongation of tapering circular bar due to an axial load P. (06 Marks)
 - c. A tension test was conducted on mild steel bar and the following data was obtained from the test:

Diameter of the bar = 18mm
 Gauge length of the bar = 82mm
 Load at proportional limit = 75kN
 Extension at a load of 62kN = 0.113mm
 Load at failure = 82kN
 Final gauge length of the bar = 106mm
 Diameter of the bar at failure = 14mm

 Determine the Young's modulus, proportional limit, true breaking stress, % elongation and percentage reduction in cross sectional area. (08 Marks)

OR

2.
 - a. Define the four elastic constants. (04 Marks)
 - b. Define principle of superposition and thermal stress. (04 Marks)
 - c. At a point in an elastic material, the stresses on two perpendicular planes are 80N/mm^2 (tensile) and 60N/mm^2 (compressive). There is also a shear stress of 40N/mm^2 . Find the normal stress and shear stress on a plane making an angle of 30° with the plane on which the tensile stress acts. Also find the values of principal stresses and the location of principal planes (adopt analytical method). (12 Marks)

Module-2

3.
 - a. Define:
 - i) Shear force
 - ii) Bending moment
 - iii) Point of contraflexure. (06 Marks)
 - b. Draw SFD and BMD for a simply supported beam subjected to a couple moment 'M' in clockwise direction acting at a distance of 'a' from left support and 'b' from right support. (06 Marks)
 - c. Sketch SFD and BMD for the beam shown in Fig.Q.3(c) indicating salient points.

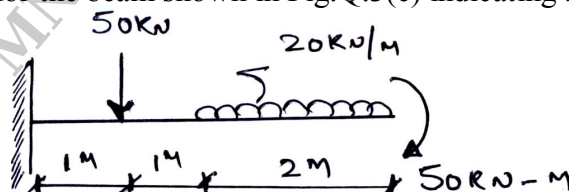


Fig.Q.3(c)

(08 Marks)

OR

- 4 a. Derive relationship between load intensity, shear force and bending moment. (06 Marks)
 b. Draw BMD and SFD for the overhanging beam shown in Fig.Q.4(b). Clearly indicate the position of contra flexure. (14 Marks)

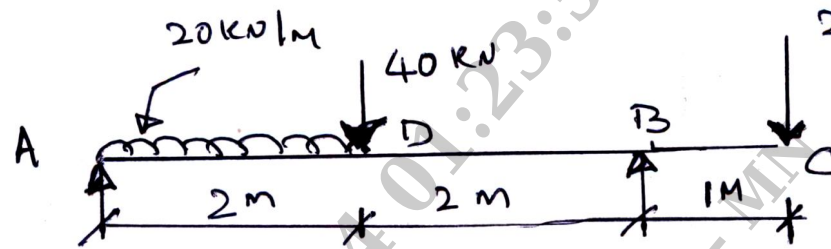
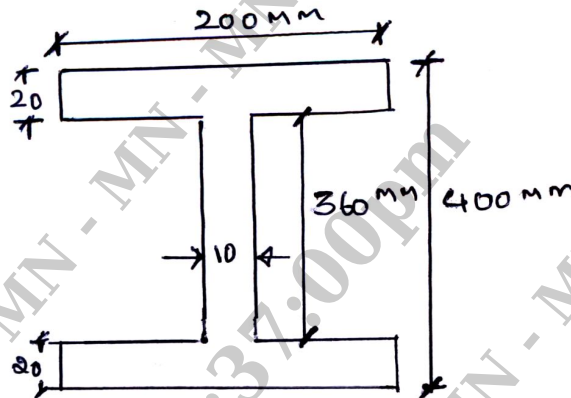


Fig.Q.4(b)

Module-3

- 5 a. List the assumptions in simple theory of bending. (04 Marks)
 b. Derive the bending equation with usual notations. (08 Marks)
 c. A rolled steel joist of I-section has the dimension as shown in Fig.Q.5(c). This beam of I-section carries a UDL of 40kN/m run on a span of 10m. Calculate the maximum stress produced due to bending. (08 Marks)



All dimensions in mm

Fig.Q.5(c)

OR

- 6 a. Define:
 i) Section Modulus
 ii) Modulus of rupture
 iii) Moment of resistance. (03 Marks)
 b. Prove that maximum shear stress in a rectangular section of width 'b' and depth 'd' is equal to 1.5 times of its average shear stress. (05 Marks)
 c. A 'T' section of flange 120mm × 12mm and overall depth 200mm with 12mm web thickness is loaded such that at a section it has a bending moment of 20kNm and shear force of 120kN. Sketch the bending and shear stress distribution diagram marking the salient values. (12 Marks)

Module-4

- 7 a. Prove that a hollow circular shaft is stiffer and stronger than a solid circular shaft in torsion which have same material length and weight, by strength and stiffness criteria. (10 Marks)
 b. Derive Lamé's equation for the radial and hoop stress for thick cylinder subjected to internal and external fluid pressure. (10 Marks)

OR

- 8 a. Determine the maximum and minimum hoop stress across the section of a pipe of 400mm internal diameter and 100mm thick, when the pipe contains a fluid at a pressure of 8N/mm^2 . Also sketch the radial pressure distribution and hoop stress distribution. (10 Marks)
- b. A shaft of hollow cross section rotates at 200rpm transmitting a power of 800kW with internal diameter = 0.8 times external diameter. Compute the diameters if the maximum shear stress is limited to 100N/mm^2 and the angle of twist to 1° in a length of 4m. Assume that the maximum torque is 30% greater than the mean torque and $G = 80\text{GPa}$. (10 Marks)

Module-5

- 9 a. Derive an expression for slope and deflection in a simply supported beam subjected to UDL throughout. Calculate the maximum slope and deflection. (06 Marks)
- b. Define: i) Buckling load ii) Effective length iii) Slenderness ratio. (06 Marks)
- c. Compute the crippling loads using Euler's and Rankine's formula for a hollow circular column 200mm external diameter and 25mm thick. The length of the column is 4m with both ends hinged. Assume $E = 200\text{GPa}$, Rankine's constants $\sigma_c = 320\text{MPa}$ and $a = 1/7500$. (08 Marks)

OR

- 10 a. Derive an equation for buckling load in a long column with both ends hinged using Euler's column theory. (08 Marks)
- b. State the assumptions made in Euler's theory. (04 Marks)
- c. A simply supported beam of length 40m is subjected to a UDL of 30kN/m over the whole span and deflects 15mm at the centre. Determine the crippling loads when this beam is used as a column with the following conditions: i) One end fixed and other end hinged ii) Both ends pin jointed [Take length of beam $l = 4000\text{mm}$ and UDL, $W = 30\text{kN/m}$, Deflection at centre = 15mm]. (08 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024 Fluid Mechanics and Hydraulics

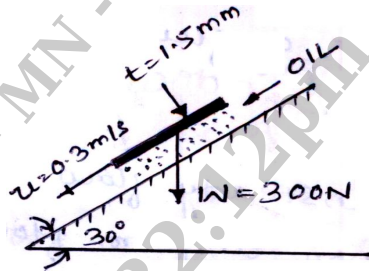
Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following fluid properties:
 - i) Density
 - ii) Specific weight
 - iii) Specific volume
 - iv) Specific gravity. (04 Marks)
- b. Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size $0.8\text{m} \times 0.8\text{m}$ and an inclined plane with angle of inclination 30° as shown in Fig.Q.1(b). The weight of the square plate is 300N and it slides down the inclined plane with a uniform velocity of 0.3m/s . The thickness of oil film is 1.5mm . (08 Marks)



- c. What are the gauge pressure and absolute pressure at a point 3m below the free surface of a liquid having a density of $1.53 \times 10^3 \text{kg/m}^3$ if the atmospheric pressure is equivalent to 750mm of mercury? The specific gravity of mercury is 13.6 and density of water $= 1000 \text{kg/m}^3$. (08 Marks)

OR

- 2 a. State and prove the Pascal's law. (06 Marks)
- b. Derive an expression for total pressure and centre of pressure on an inclined plane surface submerged in the liquid. (08 Marks)
- c. A circular plate of 3m diameter is immersed in water in such a way that its greatest and least depth below the free surface are 4m and 1.5m respectively. Determine the total pressure on the face of the plate and position of centre of pressure. (06 Marks)

Module-2

- 3 a. Derive continuity equation in Cartesian co-ordinates for a fluid in 3 dimensions. (08 Marks)
- b. Distinguish between:
 - i) Steady and unsteady flow
 - ii) Uniform and non-uniform flow
 - iii) Laminar and turbulent flow. (06 Marks)

- c. A 30cm diameter pipe, conveying water, branches into two pipes of diameters 20cm and 15cm respectively. If the average velocity in the 30cm diameter pipe is 2.5m/s, find the discharge in this pipe. Also determine the velocity in 15cm pipe if the average velocity in 20cm diameter pipe is 2m/s. (06 Marks)

OR

- 4 a. Derive the equation for discharge through venturimeter. (08 Marks)
 b. The water is flowing through a pipe having diameters 20cm and 10cm at section (1) and (2) respectively. The rate of flow through pipe is 35 litres/s. The section (1) is 6m above datum and section (2) is 4m above datum. If the pressure at section (1) is 39.24N/cm^2 , find the intensity of pressure at section (2). (06 Marks)
 c. An orifice meter with orifice diameter 15cm is inserted in a pipe of 30cm diameter. The pressure difference measured by a mercury oil differential manometer on the two sides of the orifice meter gives a reading of 50cm of mercury. Find the rate of flow of oil specific gravity 0.9 when the coefficient of discharge of the orifice meter = 0.64. (06 Marks)

Module-3

- 5 a. Give classification of orifices and mouth pieces. (06 Marks)
 b. Water discharge at the rate of 98.2 litres/s through a 120mm diameter vertical sharp edged orifice placed under a constant head of 10 meters. A point on the jet measured from the vena contracta of the jet has co-ordinates 4.5 meters horizontal and 0.54 meters vertical. Find the coefficient C_v , C_c and C_d of the orifice. (06 Marks)
 c. Derive an expression for discharge through a v-notch. (08 Marks)

OR

- 6 a. Explain major and minor losses in a flow. Give an expression for head loss due to sudden expansion in pipe line. (08 Marks)
 b. A main pipe divides into two parallel pipes which again forms one pipe. The length and diameter for the first parallel pipe are 2000m and 1m respectively, while the length and diameter of 2nd parallel pipe are 2000m and 0.8m. Find the rate of flow in each parallel pipe, if total flow in the main is $3\text{m}^3/\text{s}$. The co-efficient of friction for each parallel pipe is same and equal to 0.005 (06 Marks)
 c. Explain the phenomenon of water Hammer. List the factors upon which it depends. (06 Marks)

Module-4

- 7 a. Derive Chezy's equation for uniform rate of flow in a channel. Hence write Manning's equation. (08 Marks)
 b. The rate of flow of water through a circular channel of diameter 0.6m is 150 litres/sec. Find the slope of the bed of the channel for maximum velocity. Take $C = 60$. (06 Marks)
 c. Derive the conditions for the most economical rectangular channel. (06 Marks)

OR

- 8 a. Define hydraulic jump. List the applications of hydraulic jump. (06 Marks)
 b. The discharge of water through a rectangular channel of width 8m, is $15\text{m}^3/\text{s}$. When depth of flow of water is 1.2m. Calculate:
 i) Specific energy of the flowing water.
 ii) Critical depth and critical velocity.
 iii) Value of minimum specific energy. (06 Marks)
 c. Derive an expression for GVF in an open channel flow. (08 Marks)

Module-5

- 9 a. Derive an expression for force exerted by jet on moving curved vane in the direction of jet. (08 Marks)
- b. Explain the neat sketch, the components of a pelton wheel (impulse) turbine. (06 Marks)
- c. A jet of water of diameter 75mm moving with a velocity of 2.5m/s strikes a fixed plate in such a way that the angle between the jet and plate is 60° . Find the force exerted by the jet on the plate
- i) In the direction normal to the plate and
- ii) In the direction of the jet. (06 Marks)

OR

- 10 a. Explain with various components of a centrifugal pump. (08 Marks)
- b. The internal and external diameters of the impeller of a centrifugal pump are 200mm and 400mm respectively. The pump is running at 1200 r.p.m. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. Determine the work done by the impeller per unit weight of water. (06 Marks)
- c. Write a note of multistage centrifugal pumps. (06 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024

Analysis of Structures

Time: 3 hrs.

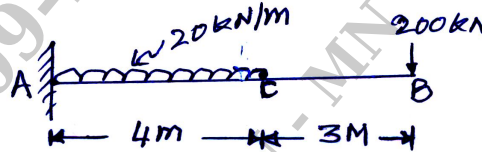
Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Any missing data assume suitably.

Module-1

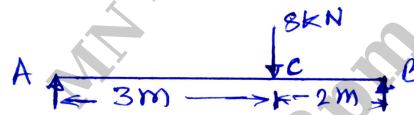
- 1 a. Determine the slope and deflection of the cantilever loaded beam shown in Fig. Q1(a) at the free end by Moment Area method. Take EI is constant. (10 Marks)

Fig. Q1(a)



- b. Determine the slope at supports and deflection under point load for the beam shown in Fig. Q1(b) using Moment Area method. Take $EI = \text{Constant}$. (10 Marks)

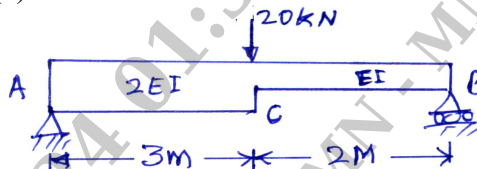
Fig. Q1(b)



OR

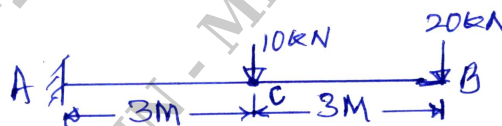
- 2 a. Determine the deflection under the point load and slope at A, using Conjugate Beam method for the shown in Fig. Q2(a). Take $EI = 2 \times 10^4 \text{ kN-m}^2$. (10 Marks)

Fig. Q2(a)



- b. Determine the slope and deflection at the free end of a cantilever beam shown in Fig. Q2(b) by Conjugate Beam method. Take $EI = 4 \times 10^4 \text{ kN-m}^2$. (10 Marks)

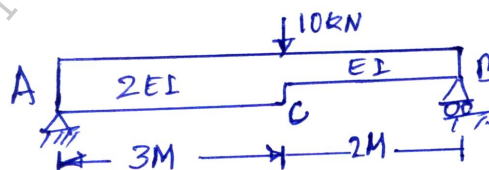
Fig. Q2(b)



Module-2

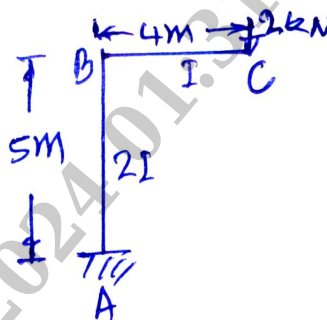
- 3 a. Determine the deflection for a simply supported beam shown in Fig. Q3(a) under the point load by Strain Energy method. Take $EI = 5000 \text{ kN-m}^2$. (10 Marks)

Fig. Q3(a)



- b. Determine the vertical deflection at 'C' in the frame shown in Fig. Q3(b) using Strain Energy method. Take $EI = 6000 \text{ kN-m}^2$. (10 Marks)

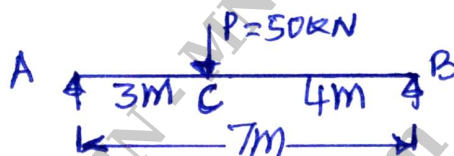
Fig. Q3(b)



OR

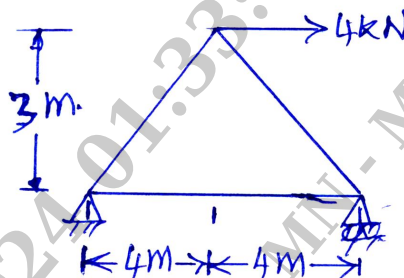
- 4 a. Find the deflection under point load for the beam shown in Fig. Q4(a) using Castigliano's First theorem. Take $E = 2800 \text{ kN-m}^2$. (10 Marks)

Fig. Q4(a)



- b. Determine the horizontal displacement of joint 'C' of the truss shown in Fig. Q4(b). The cross-sectional area of each member of the truss is $A = 400 \text{ mm}^2$, $E = 200 \text{ kN/mm}^2$. Use Castigliano's theorem. (10 Marks)

Fig. Q4(b)

**Module-3**

- 5 a. A three hinged parabolic arch has a span of 20m and rise of 5m. It carries a UDL of 25kN/m over the left half of the span and a point load of 120kN at 5m from the right end. Find the BM, Normal thrust and Radial shear at a section 4m from the left end. (10 Marks)
- b. A three hinged Segmental (circular) Arch of span 10m and central rise of 2.5m and supports a point load of 100kN at left Quarter span and Udl of 20kN/m over the right half of the span. Determine the reactions, Normal thrust and Radial shear at right quarter span. (10 Marks)

OR

- 6 a. A cable of span 20m and dip 4m carries a udl of 20kN/m over the entire span. Find the
- Maximum and Minimum tension in the cable.
 - Size and length of the cable.

(10 Marks)

- b. Determine the tension in the various segments of the cable as shown in Fig. Q6(b). Also determine the diameter of the cable required. If stress in the cable material is 150 N/mm^2 . (10 Marks)

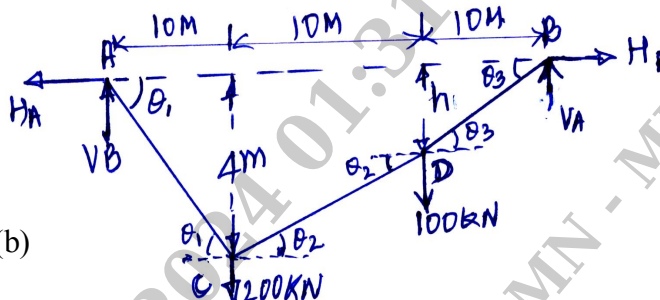


Fig. Q6(b)

Module-4

- 7 a. Analyse the two Span continuous beam as shown in Fig. Q7(a) by Slope deflection method and draw BMD. (10 Marks)

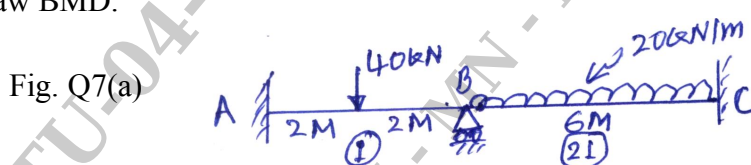


Fig. Q7(a)

- b. Analyse the beam by Slope deflection method as shown in Fig. Q7(b). The support 'B' sinks by 5m. Take $EI = 2.1 \times 10^4 \text{ kN-m}^2$. Draw BMD. (10 Marks)

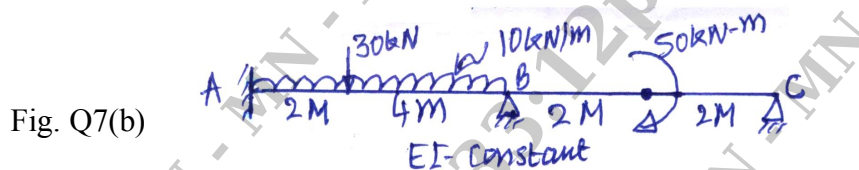


Fig. Q7(b)

OR

- 8 a. Analyse the Rigid frame by the Slope deflection method as shown in Fig. Q8(a). (10 Marks)

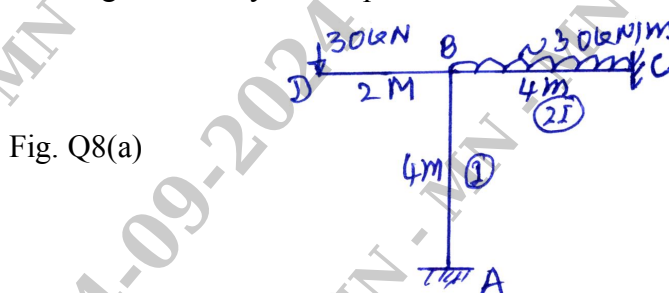


Fig. Q8(a)

- b. Analyse the Portal frame shown in Fig. 8(b) by the Slope deflection method. (10 Marks)

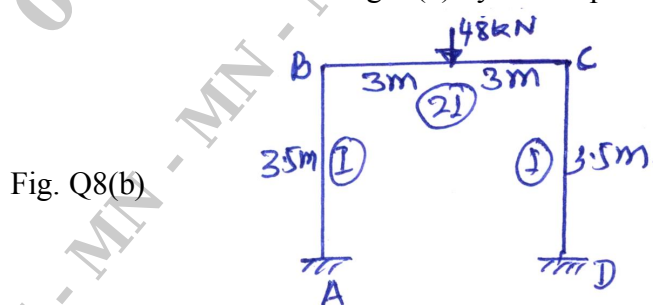
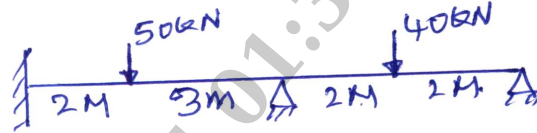


Fig. Q8(b)

Module-5

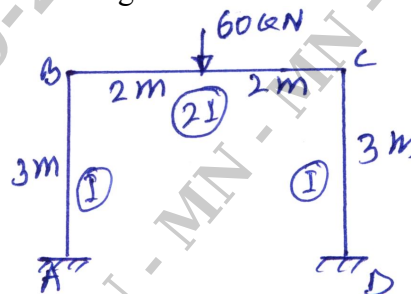
- 9 Analyse the Continuous beam shown in Fig. Q9, by Matrix flexibility method and draw BMD & SFD. Take moments as redundant. (Use system approach). (20 Marks)

Fig. Q9

**OR**

- 10 Analyse the Symmetrical frame by the Stiffness method using system approach as shown in Fig. Q10. Draw the Bending moment diagram. (20 Marks)

Fig. Q10



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CBCS SCHEME

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Green Buildings

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Soil cement blocks help in reducing the amount of :
a) Sand b) Cement c) Water d) Mortar
 2. Stone blocks help in saving materials by reducing wall thickness from :
a) 30 – 20mm b) 20 – 10 cm c) 20 – 10 mm d) 30 – 20 cm
 3. FRP stands for
a) Fibre reinforced polymer
b) Fully reinforced polymer
c) Fire resistant polymer
d) Fibre reconditioned polymer
 4. In _____ cement mortar is reinforced with wire meshes
a) Fibre reinforced cement
b) Wire reinforced cement
c) Ferrocement
d) Mesh mortar
 5. Fibre Reinforced Concrete (FRC) is used in :
a) Pavements b) Foundation c) Slab d) Beam

6. Concrete hollow bricks have the following characteristics :
- a) Less thermal insulation
 - b) Costly
 - c) Fast construction
 - d) Decrease floor area
7. Which of the following is/are basic building materials?
- a) Wood
 - b) Cement
 - c) Brick
 - d) All of these
8. Lime replaces the use of cement by 50% in lime based stone masonry.
- a) True
 - b) False
9. Recycling steel reduces
- a) Energy use by 75%
 - b) Energy use by 50%
 - c) Energy use by 35%
 - d) Energy use by 20%
10. Which of the below roofing materials is most effective?
- a) Asphalt
 - b) Plastic
 - c) Rubber
 - d) Metal
11. In Rat_trap bond masonry course height is
- a) 110mm
 - b) 75mm
 - c) 230mm
 - d) 190mm
12. By using filler slab technique :
- a) 20% saving of concrete is there
 - b) Aesthetic ceiling is obtained
 - c) Less comfort is there
 - d) Load on foundation in creases
13. Economy can be achieved indoors, windows, ventilator by :
- a) No frame
 - b) Less width
 - c) Steel frames
 - d) Less height
14. For sloping roofs, ferro cement roofing units can be used
- a) True
 - b) False
15. Rat – trap bond masonry requires _____ less mortar
- a) 10%
 - b) 50%
 - c) 40%
 - d) 20%
16. Which of the below is not a filler material in fillar slabs?
- a) Rubber pieces
 - b) clay pots
 - c) Burnt clay bricks
 - d) Coconut shell

17. For natural and cost-effective ventilation, _____ can be used
a) Hollow bricks b) Wall openings c) Skylight d) Brick Jali
18. COSTFORD full form
a) Centre of science and technology for rural development
b) Centre of science and technology for rural deployment
c) Centre of science and technology for rural department
d) Centre of science and technology for rural division
19. The agency which demonstrate the practical usage of cost effective technology and housing concepts through model houses
a) COSTFORD
b) NIRMITHI KENDRA
c) Habitat
d) All of these
20. The Nirmithi Kendras are established under the chairmanship of
a) Deputy Commissioner of District
b) Chief Executive Officer, Zilla Panchayat
c) District Commissioner
d) MP of District
21. Which one of the following cause global warming?
a) Carbon dioxide b) Oxygen c) Nitrogen d) Hydrogen
22. Which one of the following result takes place due to global warming?
a) Maintaining steady temperature
b) Changes in rainfall
c) Pleasant environment
d) Causing less pollution
23. Which one of the following cause global warming?
a) Radiative forcing
b) Earth gravitation force
c) Oxygen
d) Centripetal force
24. The concept of carbon foot print was developed by?
a) UNFCC
b) UNEP
c) William E. Rees and Mathis Wackernagel
d) Ernst Haeckel

25. According to carbon footprint challenge, India will slash its carbon emission by _____ billion tons by the year 2030
a) 1 b) 2 c) 3 d) 4
26. You can help to decrease your carbon footprint by _____
a) Buying daily use stuff from online
b) By eating non-veg foods
c) Buying your needs from the local shop
d) Travelling a lots
27. Efforts taken in India to reduce global warming
a) Focus on renewable energy
b) Mobilisation of resources
c) International of Solar Alliance (ISA)
d) All of these
28. Embodied energy measured in buildings as
a) Initial embodied energy
b) Recurring embodied energy
c) Operational energy
d) All of these
29. The construction cost of green industrial manufacturing building is about _____ % higher than that of a conventional building
a) 18% b) 28% c) 38% d) 48%
30. The green building uses comparatively _____ % lower energy than
a) 11% b) 21% c) 31% d) 41%
31. Which of the following green rating systems are currently working in India?
a) LEED b) GRIHA c) both a and b d) none of these
32. LEED means _____
a) Leadership in energy and efficiency design
b) Leadership in energy and efficiency document
c) Leadership in energy and environmental design
d) Leadership in energy and Environmental document
33. LEED gives rating in form
a) 1star, 2 star, 3 star, 4 star, 5 star
b) Platinum, Gold and Sliver
c) both a and b
d) none of these

34. GRIHA means that m
a) Green Rating for Indian Habitat Assessment
b) Green Rating for Integrated Habitat Assessment
c) Green Rating for International Habitat Assessment
d) Green Rating for Information Habitat Assessment
35. When GRIHA was launched in India?
a) 2018 b) 2006 c) 2010 d) 2011
36. Sustainable planning considers, environmental, social, and _____ impacts of a buildings
a) Technological b) Economical c) Both a and b d) None of these
37. Which of the following options is not included in the sustainable development parameters?
a) Gender inequality and diversity
b) Intergenerational equality
c) Growing annually
d) None of these
38. What is the definition of sustainable development?
a) The growth that satisfies current demands without jeopardizing future generations ability to fulfill their own needs
b) Conserve mineral wealth and explore alternative energy sources while decreasing pollution and environmental impact
c) It is the process of creating land and building projects in such a way that they have environmental effect by enabling them to produce fuel efficient self sufficiency patterns
d) All of these
39. Sustainable building means that
a) Green building
b) Environmental building
c) Both “a” and “b”
d) none of these
40. Characteristics of Green buildings
a) location and transport
b) sustainable sites
c) efficient use of water
d) All of these
41. Solar energy knows as “_____”
a) power energy
b) green energy
c) electrical energy
d) all of these

42. Reasons use solar technologies for buildings
a) Increasing energy consumption
b) Environmental consciousness
c) Global climate change
d) All of these
43. Solar powered building improve that heating and cooling system's efficiency by _____ %
a) 30 b) 20 c) 10 d) 0
44. The key components of water efficiency in green building according to the USGBC are :
a) Reduce indoor potable water use
b) Reduce water consumption to save energy
c) Improve Environmental well being
d) All of these
45. The methods to water management are as follows
a) Rain water harvesting
b) Ground water recharge
c) Artificial ground water recharge
d) Drip irrigation
46. How many major sources of solids waste are there based on their origin?
a) 10 b) 5 c) 9 d) 6
47. The term ISWM refers to
a) International Solid Waste Management
b) Integrated Solid Waste Management
c) International Solid Waste Machine
d) integrated Solid Waste Machine
48. In how many stages the treatment of waste-water in STPs is carried out?
a) One b) Two c) Three d) Four
49. Green cover is a natural or planted vegetation covering a certain area of terrain, functioning as protection against soil erosion, protecting the fauna, and balancing the temperature
a) True b) False
50. Built environment can be defined as
a) human made surroundings that provide the setting for human activity
b) material, spatial and cultural product of human labor that combines physical elements and energy in forms of living working and playing
c) human made space in which people live, work and recreate on a day to day basis
d) All of these.

CBCS SCHEME

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21CV51

Fifth Semester B.E. Degree Examination, June/July 2024 Hydrology and Water Resources Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. Explain Horton's hydrological cycle with sketch. (08 Marks)
 - b. Discuss the methods of computation of average rainfall over a basin. (06 Marks)
 - c. A catchment has five rain gauge stations. In a year the annual rainfall recorded by the gauges are 78.8cm, 90.2cm, 98.6cm, 102.4cm and 70.4cm for a 6% error in the estimation of the mean rainfall, determine the additional number of rain gauges needed. (06 Marks)

OR

- 2
 - a. With a neat sketch explain Indian standard evaporation pan. (06 Marks)
 - b. Explain infiltration capacity curve with sketch. (06 Marks)
 - c. The annual rainfall at station x and the average of annual rainfall at 10 surrounding base stations in cm are given below for a period of 16 years starting from 1979.

Year	1970	1971	1972	1973	1974	1975	1976	1977
Rainfall @x (cm)	95	75	90	85	95	90	100	90
10 station average (cm)	75	55	70	65	75	70	70	70
Year	1978	1979	1980	1981	1982	1983	1984	1985
Rainfall @x (cm)	75	65	70	87	67	62	55	69
10 station average (cm)	75	65	90	86	68	67	62	77

(08 Marks)

Module-2

- 3
 - a. Discuss the runoff and explain the factors affecting runoff. (06 Marks)
 - b. With a neat sketch explain the components of a single peaked hydrograph. (06 Marks)
 - c. The rate of rainfall for successive 30 min periods of a 4 hour storm are given below :
3.5, 6.5, 8.5, 7.8, 6.4, 4.0, 4.0, 6.0 cm/hr. Take a value of ϕ - index as 4.5cm/hr compute the following :
 i) total rainfall ii) total rainfall excess iii) W_i . (08 Marks)

OR

- 4
 - a. Define unit hydrograph, and construction of UH. (06 Marks)
 - b. Describe the methods separation of base flow and direct runoff in a single peaked storm hydrograph. (06 Marks)
 - c. Find the ordinates of storm hydrograph resulting from a 3 hr storm with rainfall of 2, 6.75 and 3.75cm during subsequent 3 hr intervals. The ordinates of 3 hr hydrograph are given below :

Hours	03	06	09	12	15	18	21	24	03	06	09	12	15	18	21	24
UGO cumes	0	110	365	501	390	310	250	235	175	130	95	69	40	22	10	0

(08 Marks)

Module-3

- 5 a. Explain the irrigation methods with sketches. (06 Marks)
 b. Explain drip irrigation with sketches. (06 Marks)
 c. A loam soil has field capacity of 22% and wilting coefficient of 10%. The dry unit weight of soil is 15kN/m^3 . If the root zone depth is 70cm, determine the storage capacity of the soil. Irrigation water is applied when moisture content falls to 14%. If the water application efficiency is 75%. Determine the water depth required to be applied in the field. (08 Marks)

OR

- 6 a. Explain irrigation efficiencies with equations. (06 Marks)
 b. Define duty, delta and derive the relation between them. (06 Marks)
 c. The base period intensity of irrigation and duty of various crops under a canal system are given in the table below. Find the reservoir capacity of the canal losses are 20% and reservoir losses are 12%.

Crop	Base period days	Duty at the field hr/cume	Area under the crop(hect)
Wheat	120	1800	4800
Sugar cane	360	800	5600
Cotton	200	1400	2400
Rice	120	900	3200
Vegetable	120	200	1400

(08 Marks)

Module-4

- 7 a. What are the types of canals explain with sketches. (08 Marks)
 b. Explain Kennedy's method of canal design. (06 Marks)
 c. Explain zones of storage in reservoir with sketch. (06 Marks)

OR

- 8 a. How the reservoir capacity is calculated for a specified yield from the mass inflow curve. (08 Marks)
 b. Discuss the factors affecting the site for selection of a reservoir. (06 Marks)
 c. The construction cost for certain possible height of a dam at a given site have been estimated and are tabulated. The storage capacity for all these dam heights are also given. Determine the most economic height of the dam.

Height of dam in (mt)	10	20	30	40	50	60	70
Construction cost (million Rs)	4	8	12	18	27	39	50
Storage in (mcm)	50	110	180	250	350	500	600

(06 Marks)

Module-5

- 9 a. Define marginal bund and design steps for marginal bunds. (08 Marks)
 b. What are the causes of flood? Explain any five. (06 Marks)
 c. Discuss the measures for water conservations and augmentation. (06 Marks)

OR

- 10 a. Explain rain water harvesting methods. (08 Marks)
 b. How the restoration and rejuvenation of old tanks and carried out. (06 Marks)
 c. What are drought management schemes in India? (06 Marks)

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21CV52

Fifth Semester B.E. Degree Examination, June/July 2024 Transportation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. How roads are classified according to Nagpur Road Plan explain functions of each category of road. (10 Marks)
- b. Explain basic requirements of an idea alignment and explain the various factors controlling the Highway alignment. (10 Marks)

OR

- 2 a. What are the different modes of transportation? Explain briefly. (05 Marks)
- b. Explain different road patterns with sketch. (05 Marks)
- c. The area at a certain district in India is 14000km² and there are 10 towns as per 1981 census. Determine the length of different categories of roads to be provided in this district by year 2001. (10 Marks)

Module-2

- 3 a. Sketch the component layers of flexible pavement. Explain the function of each component layer. (10 Marks)
- b. Determine the height of crown for a MDR of width 3.8m with WBM pavement. The rainfall is light in that area. (05 Marks)
- c. Explain PIEV theory with sketch. (05 Marks)

OR

- 4 a. Design rate of super elevation for 100 kmph speed having 500m radius. (05 Marks)
- b. Calculate safe stopping sight distance for 50kmph speed. Assume remaining data as per IRC specification. (05 Marks)
- c. Explain the procedure of finding Equivalent Single Wheel Load (ESWL) by graphical method. (10 Marks)

Module-3

- 5 a. Explain the procedure of finding crushing value and impact value of aggregates. (10 Marks)
- b. The load penetration values of CBR test conducted on specimen of soil are given below. Determine CBR value of soil if 100 division = 190kg.

Penetration of plunger (mm)	0.0	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0	7.5	10	12.5
Load readings (division)	0	8	15	23	29	34	37	43	48	57	63	67

(10 Marks)

OR

- 6 a. Calculate the group index of a sample of soil with the following data :
i) Percentage passing in 75 micron sieve = 60
ii) Liquid limit = 30 plasticity index = 12. (10 Marks)
b. Describe procedure of finding penetration value of Bitumen sample and softening point of bitumen sample. (10 Marks)

Module-4

- 7 a. Elaborate BOOT concept. (05 Marks)
b. Discuss importance of highway drainage. (05 Marks)
c. List and explain the highway user benefits. (10 Marks)

OR

- 8 a. Differentiate and explain surface and sub-surface drainage system. (10 Marks)
b. List and explain different methods of highway economic analysis. (10 Marks)

Module-5

- 9 a. Draw the sketch of coning of wheel and canting of rail. Explain the same. (10 Marks)
b. List and explain the Aircraft characteristics. (10 Marks)

OR

- 10 a. Brief the Traffic regulations and control. (05 Marks)
b. Explain briefly the traffic characteristics. (05 Marks)
c. Explain orientation of runway by using wind rose diagram. (10 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024 Design of RC Structural Elements

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS456-2000, SP-16 is permitted.
3. Assume any missing data suitably.

Module-1

- 1 a. Explain the difference between working stress and limit state method of design. (08 Marks)
- b. Write brief short notes on:
 - i) Partial safety factor for loads
 - ii) Partial safety factor for material
 - iii) Characteristic load
 - iv) Characteristic strength. (12 Marks)

OR

- 2 a. Explain the terms balanced section, under reinforced and over reinforced section. (06 Marks)
- b. A simply supported beam of rectangular cross section 250mm × 450mm overall depth is used over an effective span of 4.0m. The beam is reinforced with 3 bars of 20mm diameter, Fe415 HYSD bars at an effective depth of 400mm. Two anchor bars of 10mm diameter bars are provided. The self weight of beam together with the dead load on the beam is 4kN/m service load on the beam is 10kN/m. Use M20 grade of concrete. Compute: i) Short term deflection ii) Long term deflection. (14 Marks)

Module-2

- 3 a. Derive from fundamentals the expression for the area of stress block $0.36f_{ck} x_u$. (08 Marks)
- b. A rectangular R.C. beam section 230mm × 500mm is used as a simply supported beam for an effective span of 6.0m. The beam consists of tensile reinforcement of 4000mm² and centre of the reinforcement is placed at 35mm from bottom edge. What total maximum udl can be allowed on the beam? Use M20 concrete and Fe415 grade steel. (12 Marks)

OR

- 4 a. Derive the moment of resistance equation for a singly reinforced rectangular section. (10 Marks)
- b. A RCC beam 250mm wide and 450mm deep is reinforced with 3nos of 20mm diameter bars of grade Fe-415 on the tension side with an effective cover of 50mm. If the shear reinforcement of 2 legged 8mm vertical stirrups at a spacing of 160mm centre to centre is provide at a section. Determine the design shear strength of the section. Assume M-20 concrete has been used. (10 Marks)

Module-3

- 5 Design a reinforced concrete beam of rectangular section using the following data.
Effective span = 5m, width of beam = 250mm, overall depth = 500mm service load including dead load and live load = 40kN/m, effective cover = 50mm. Adopt M20 concrete and Fe-415 grade steel. Sketch the reinforcement details. (20 Marks)

OR

- 6 A T-beam slab floor has 125mm thick slab forming a part of tee beam. Which are 8.0m clear span? The end bearings are 450mm thick. Spacing of T beam is 3.5m c/c. The live load on the floor is 3kN/m^2 . Design one of the intermediate T beams. Use M-20 concrete and Fe-415 grade steel. (20 Marks)

Module-4

- 7 Design a reinforced concrete slab for a room of size $3\text{m} \times 5\text{m}$ (clear). The slab is supported on a wall of 300mm thickness with corners held down. Two adjacent edges of the slab are continuous and other two edges discontinuous. The live load on slab is 3kN/m^2 . Assume floor finish of 1kN/m^2 . Use M20 grade concrete and Fe-415 grade steel sketch the reinforcement details. (20 Marks)

OR

- 8 Design a dog legged stair for an office building in a room measuring $2.8\text{m} \times 5.8\text{m}$ clear vertical distance between the floor is 3.6m. The width of flight is to be 1.25m. Assume live load of 3kN/m^2 . Use M-20 concrete and Fe-415 grade steel. Assume the stairs are supported on 230mm at the outer edges of landing stairs. Sketch the reinforcement details. (20 Marks)

Module-5

- 9 a. Design a short column of size $450\text{mm} \times 600\text{mm}$. The column is subjected to a factored load of 3000kN. Use M20 concrete and Fe415 grade steel. (10 Marks)
b. Design the reinforcement for a column of size $300\text{mm} \times 400\text{mm}$ having an effective length of 2.5m. Moment about the major axis of the column is 100kN-m and axial load of 800kN. Use M25 concrete and Fe500 steel provide the reinforcement on two sides. (10 Marks)

OR

- 10 Design a square footing to carry a column load of 1200kN from a 400mm square column. The bearing capacity of soil is 120kN/m^2 . Use M20 concrete and Fe-415 grade steel. Sketch the reinforcement details. (20 Marks)

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21CV54

Fifth Semester B.E. Degree Examination, June/July 2024 Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of phase diagram, define the following terms:
i) Void Ratio ii) Saturated density iii) Degree of saturation iv) Moisture content. (08 Marks)
- b. Describe the method of determining liquid limit of soil by Casagrande's method. (08 Marks)
- c. Explain the classification of fine grained soil using plasticity chart. (04 Marks)

OR

- 2 a. Define the following:
i) Specific gravity of soil.
ii) Relative density
iii) Liquid limit of soil
iv) Plastic limit of soil. (08 Marks)
- b. With usual notations prove that $e_s = W_G$. (04 Marks)
- c. Explain the procedure of determining water content of soil by oven drying method. (08 Marks)

Module-2

- 3 a. List and explain the factors affecting permeability of soil. (08 Marks)
- b. Differentiate between discharge velocity and seepage velocity. (04 Marks)
- c. A soil deposit 8m thick has bulk unit weight of 20 kN/m^3 . If the water table is at a depth of 3m below ground surface and soil above water table is saturated by capillary water. Find total, effective and neutral stress @ 8m, 3m and 0m depth. (08 Marks)

OR

- 4 a. Describe an experiment to determine permeability of clayey soil with a neat sketch. (08 Marks)
- b. Write a short notes on: Quick sand phenomenon. (04 Marks)
- c. A clay strata 6m thick is laying below a sand layer 5m thick. The water table is at a depth of 2m from the surface, the sand has a porosity of 40% and $G = 2.7$. Sand above the water table may be taken as a dry. The water content of soil 60% and $G = 2.65$. Calculate the total stress, pore water pressure and effective stress at the middle of the clay layer. (08 Marks)

Module-3

- 5 a. List and explain the factors affecting compaction property of soil. (08 Marks)
- b. Explain the procedure to find pre consolidation pressure. (06 Marks)
- c. Give step by step procedure of finding compaction properties through standard proctor test. (06 Marks)

OR

- 6 a. List the assumptions made in Terzaghi's one dimensional consolidation theory. (08 Marks)
 b. Explain the effect of compaction on soil properties. (06 Marks)
 c. Describe laboratory one dimensional consolidation test. (06 Marks)

Module-4

- 7 a. Explain consolidation process through mass-spring analogy. (10 Marks)
 b. With neat sketch explain direct shear test. (10 Marks)

OR

- 8 a. With neat sketch explain triaxial shear test. (10 Marks)
 b. List and explain the factors affecting shear strength of soil. (10 Marks)

Module-5

- 9 a. Explain Terzaghi's method to determine the bearing capacity of soil. (10 Marks)
 b. A strip footing 2.5m wide at its base is located at a depth of 1.5m below the ground surface. The properties of the foundation soil are : $\gamma = 18\text{kN/m}^3$, $\phi = 20^\circ$ and $C = 30\text{kN/m}^2$. Determine the safe bearing capacity, using a factor of safety 3, if the water table is at a depth of 1m below the ground surface and at the ground surface. Use Terzaghi's analysis. Take $N_C = 17.7$, $N_q = 7.4$, $N_\gamma = 5.0$. (10 Marks)

OR

- 10 a. Explain the different modes of shear failure with neat sketch. (10 Marks)
 b. A 3.0m square footing is located in a dense sand at a depth of 2.0m. Determine ultimate bearing capacity for the following water table positions:
 i) At the ground surface
 ii) At footing level
 iii) At 1m below the footing.
 The moist unit weight of sand above the water table is 18kN/m^3 and the saturated weight is 20kN/m^3 , $\phi = 35^\circ$ and $C = 0$. Take $N_q = 33$ and $N_\gamma = 34.0$. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Construction Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Management. Explain the characteristics of Management. (08 Marks)
- b. Explain the functional areas of Management. (08 Marks)
- c. Discuss the dependencies in a Gantt chart, with neat sketches. (04 Marks)

OR

- 2 a. With illustration, explain AoA and AoN Network diagrams. (08 Marks)
- b. A Project consists of the following activities. Draw the network diagram, calculate EST, EFT, LST, LFT, FT and FF. (08 Marks)

Activity	10-20	10-30	20-40	30-40	20-50	40-50
Duration (days)	13	12	2	8	15	2

- c. What is Work Break down structure? Mention its uses. (04 Marks)

Module-2

- 3 a. What are the factors affecting Labor Productivity in Construction Industry? Briefly explain. (08 Marks)
- b. Calculate time required to grade and finish 50 kms of road formation with width equal to thrice the width of the motor grader, using six passes of the motor grader with speed for each of the successive two passes as 6 km/h, 8km/h and 10km/h respectively. Assume machine efficiency based on operation skill, machine characteristic and working condition as 80%. (08 Marks)
- c. Discuss the class of labour in Construction Project. (04 Marks)

OR

- 4 a. Briefly explain the points to be considered for selection of Construction Equipments. (08 Marks)
- b. A Company has purchased an equipment of Rs 1,50,000 with an estimated life of 10 years. The estimated salvage value of the equipment at the end of its lifetime is Rs 25,000/-. Determine the depreciation charge and book value at the end of various years using the decline balance method of depreciation by assuming 0.2 for K. (08 Marks)
- c. List out the functions of Materials Management. (04 Marks)

Module-3

- 5 a. Explain the process of Project Quality Management. (10 Marks)
- b. Explain the safety measures adopted during drilling and blasting. (10 Marks)

OR

- 6 a. Discuss in detail about the cost of Quality in Construction. (10 Marks)
- b. Explain the types of Conflict of Interest. (06 Marks)
- c. Discuss the following : (04 Marks)
 - i) Gift and Bribes
 - ii) Whistle Blowing.

Module-4

- 7 a. What is Engineering Economy? Explain the principles of Engineering Economics. (08 Marks)
 b. An engineer has two bids for an excavator to be installed in a new building. The detail of the bids for the excavator are as follows :

Bid	Engineer's Estimate		
	Initial Cost (Rs)	Service Life (Years)	Annual Operation Maintenance Cost
Company A	10,50,000	15	60,000
Company B	11,00,000	15	70,500

Determine which bid should be accepted based on the present worth method of comparison assuming 18% interest rate, compounded annually. (12 Marks)

OR

- 8 a. Explain Break Even Analysis. Mention the assumptions of Break Even Analysis. (08 Marks)
 b. Determine the effective interest rate for a nominal annual rate of 8% that is compounded.
 i) Daily ii) Monthly iii) Quarterly iv) Semi annually. (12 Marks)

Module-5

- 9 a. Discuss in detail about the Project report for starting a new venture. (10 Marks)
 b. Explain the role and significance of Venture Capital. (10 Marks)

OR

- 10 a. Explain different types of feasibility study carried out to start business. (10 Marks)
 b. What are the different ways to entry into International business? Explain any two ways. (10 Marks)

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21CV62

Sixth Semester B.E. Degree Examination, June/July 2024 Concrete Technology

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of Code book IS – 10262 – 2019 is permitted.

Module-1

- 1 a. Explain the constituents of cement with their percentage and their function. (08 Marks)
- b. List out Bogues compounds C_2S , C_3S , C_3A and C_2AF . Explain their contribution towards gaining of strength of cement. (08 Marks)
- c. Describe the process of hydration of cement. (04 Marks)

OR

- 2 a. Explain the manufacturing process of cement by dry process along with the flow chart. (08 Marks)
- b. List the types of cement and briefly explain the properties and application of any four types of cement. (08 Marks)
- c. Explain the importance of size, shape and texture of aggregate. (04 Marks)

Module-2

- 3 a. Explain the factors affecting the workability of fresh concrete. (10 Marks)
- b. Mention various stages involved in manufacturing of discuss any three. (10 Marks)

OR

- 4 a. Define workability. How do you measure the workability of the concrete by slump cone apparatus with a neat sketch? (10 Marks)
- b. Why curing is needed? Explain different methods of curing of concrete in detail. (10 Marks)

Module-3

- 5 a. Define admixture. Briefly explain the classification of admixture. (10 Marks)
- b. Explain the effect of superplasticizer and accelerators on the properties of fresh and hardened concrete. (10 Marks)

OR

- 6 Design a concrete mix for grade M25
- | | |
|---|-------------------------|
| a. Grade designation | : M25 |
| b. Type of cement | : OPC 43 grade |
| c. Maximum nominal size of aggregate | : 20mm down |
| d. Minimum cement content | : 300kg/m ³ |
| e. Water cement ratio | : 0.5 |
| f. Workability | : 75mm slump |
| g. No chemical admixture | |
| h. Fine aggregate | : Zone . II |
| i. Exposure condition | : Moderate |
| j. Method of concrete placing | : Manual |
| k. Maximum cement content | : 450 kg/m ³ |
| l. Specific gravity of cement | : 3.15 |
| m. Specific gravity of coarse aggregate | : 2.80 |
| n. Water absorption of coarse aggregate | : 1% |
| o. Free surface moisture | : Nil |
| p. Specific gravity of fine aggregate | : 2.65 |
| q. Water absorption of fine aggregate | : 2% |
| r. Free surface moisture | : 2%. |

(20 Marks)

Module-4

- 7 a. Explain the factors influencing the strength of concrete. (10 Marks)
 b. What is maturity of concrete? Explain its significance in gaining the strength of concrete. (10 Marks)

OR

- 8 a. Explain the factors affecting modulus of elasticity of concrete. (10 Marks)
 b. Explain flexural strength and split tensile strength lists on concrete. (10 Marks)

Module-5

- 9 a. What is durability of concrete? What are the factors affecting durability of concrete. (10 Marks)
 b. Define shrinkage and creep of concrete. Discuss about the factors affecting shrinkage of concrete. (10 Marks)

OR

- 10 a. Explain in brief chloride and sulphate attack on concrete and its affect on durability of concrete. (10 Marks)
 b. Explain the process of carbonation, freezing and thawing in concrete. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Design of Steel Structure

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS800-2007, Handbook SPC(Part-1), 1964 is permitted.

Module-1

- 1 a. Describe briefly advantages and disadvantages of steel structures. (06 Marks)
- b. State the upper bound, lower bound and uniqueness theorems. (06 Marks)
- c. Explain the four classes of section as per IS 800:2007. (08 Marks)

OR

- 2 a. Explain the load combinations adopted in the design of steel structures. (06 Marks)
- b. Determine the shape factor of a 'T' section having a flange width 100 mm and 10 mm thick with a web of 10 mm thick and 150 mm depth. (08 Marks)
- c. Define : (i) Plastic hinge (ii) Mechanism (iii) Load factor (06 Marks)

Module-2

- 3 a. Explain the various modes of failure of bolted connections with neat sketches. (06 Marks)
- b. Describe briefly advantages and disadvantages of welded connections. (06 Marks)
- c. A tie member in a truss girder is 250mm×14mm in size. It is welded to a 10 mm thick gusset plate by a fillet weld. The overlap of the member is 300 mm and the weld size is 6 mm. Determine the design strength of the joint, if the welding is done as shown in Fig. Q3 (c). What is the increase in strength of the joint, if welding is done allround. Assume shop welding. (08 Marks)

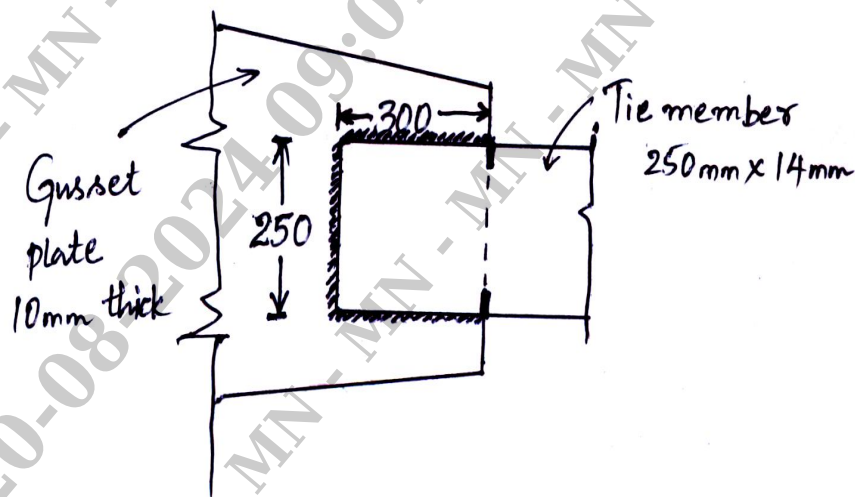


Fig. Q3 (c)

OR

- 4 a. What are the requirements of an ideal welded joint? (06 Marks)

- b. Determine the safe load 'P' that can be carried by the joint shown in Fig.4 (b). The bolts used are 20 mm diameter of grade 4.6. The thickness of the flange of I-section is 9.1 mm and that of bracket plate 10 mm. (14 Marks)

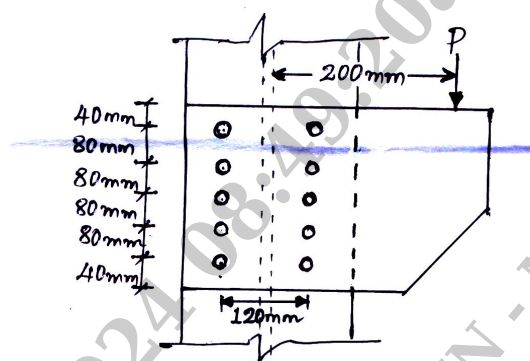


Fig. Q4 (b)

Module-3

- 5 a. Explain the factors affecting strength of a column. (06 Marks)
 b. A column 5 m long has to support a factored load of 5000 kN. It is effectively held in position and direction at both ends. Design the column using ISHB400@0.806 kN/m and 20 mm cover plates. (Both the ends are fixed). (14 Marks)

OR

- 6 a. Mention design specifications for battened system as per IS800-2007. (04 Marks)
 b. Explain the different failure modes of compression members. (06 Marks)
 c. Design a rolled steel beam section column to carry an axial load of 1000 kN at service condition. The length of column is 4 m. Both ends are hinged. (10 Marks)

Module-4

- 7 a. Explain the following :
 (i) Lug angles.
 (ii) Shear lag. (06 Marks)
 b. Design a tension member 3.4 m between center to center of intersection using double angle section carrying a factored load of 700 kN. (14 Marks)

OR

- 8 a. Explain the components of a gusseted base with neat sketch. (06 Marks)
 b. A column section ISHB350@661.2 N/m carries a factored axial compressive load of 1650 kN and factored bending moment of 90 kN-m. Design the base plate. (14 Marks)

Module-5

- 9 a. Explain the factors affecting the lateral stability of beams. (08 Marks)
 b. Explain web crippling and web buckling in flexural member with sketches. (06 Marks)
 c. Explain briefly shear strength of steel beams. (06 Marks)

OR

- 10 a. Explain briefly lateral unsupported beams. (04 Marks)
 b. A simply supported steel joist of 4.0 m effective span is laterally supported throughout. It carries a total uniformly distributed load of 40 kN. (Inclusive of self weight). Design a suitable steel section of grade Fe410. (16 Marks)

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21CV646

Sixth Semester B.E. Degree Examination, June/July 2024 Alternative Building Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain energy in building materials. Illustrate content in one burnt brick. (10 Marks)
b. List the four categories of energy consumption in building and explain them. (10 Marks)

OR

- 2 a. List out various environmental friendly and cost effective building technologies. Explain any one in brief. (10 Marks)
b. What are the advantages of LEED? List out main categories in LEED rating system. (10 Marks)

Module-2

- 3 a. Explain table moulded bricks and wire cut brick along with characteristics. (10 Marks)
b. Write short notes on: (i) Late rite blocks (ii) Fal G blocks (10 Marks)

OR

- 4 a. Explain concept of Stabilized Mual Block (SMB). (10 Marks)
b. List the properties of good mortar and explain any two in detail. (10 Marks)

Module-3

- 5 a. Name different types of pozzolona material. Explain any two in detail. (10 Marks)
b. Write notes on: (i) Metallic fiber (ii) Natural fiber (10 Marks)

OR

- 6 a. List the agro wastes and mention its application in the building construction. (10 Marks)
b. What are different industrial wastes used effectively in construction industry? Explain any two materials. (10 Marks)

Module-4

- 7 a. What are primary functions of a roof? Explain briefly the various roofing alternative. (10 Marks)
b. What are the materials used in ferrocement? Explain the construction method in brief. (10 Marks)

OR

- 8 a. Explain composite masonry and cavity wall masonry. (10 Marks)
b. Write the advantages and disadvantages of Mivan construction. (10 Marks)

Module-5

- 9 a. What are the advantages and disadvantages of manufacturing concrete from RMC plant? (10 Marks)
b. What is meant by precast elements? What are the advantages of precast elements? (10 Marks)

OR

- 10 a. Write short notes on: (i) Type of concrete mixers (ii) Cost concept in buildings (10 Marks)
b. What are the equipments used for producing stabilized mud blocks? Explain in brief. (10 Marks)

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BESCK104A/BESCKA104

First Semester B.E./B.Tech. Degree Examination, June/July 2024

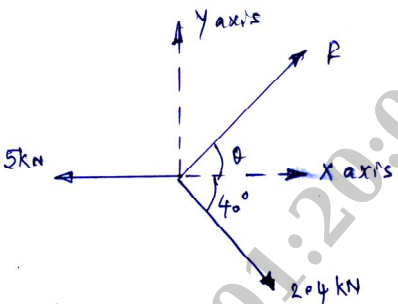
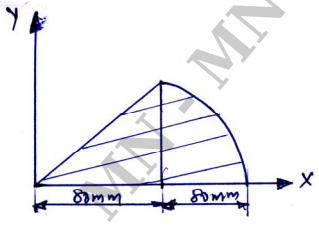
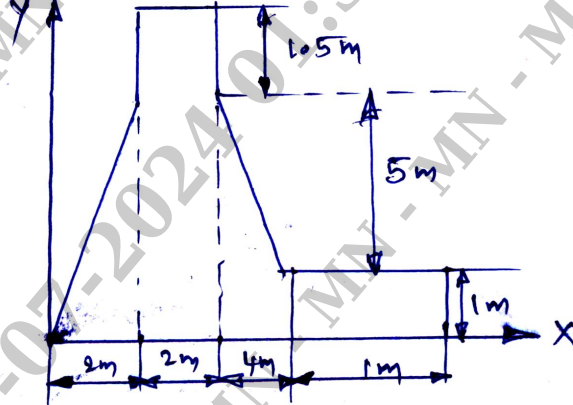
Introduction to Civil Engineering

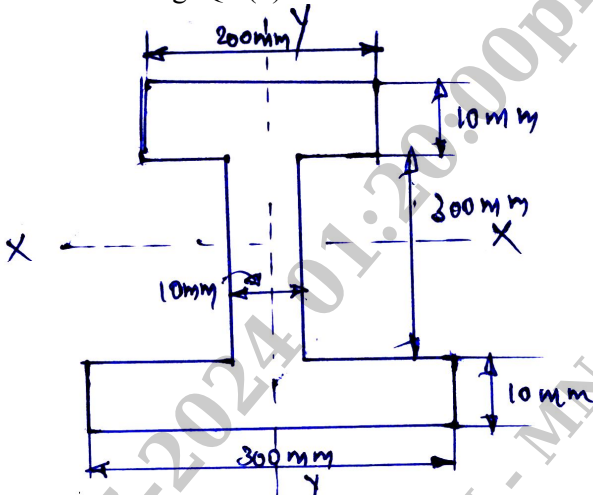
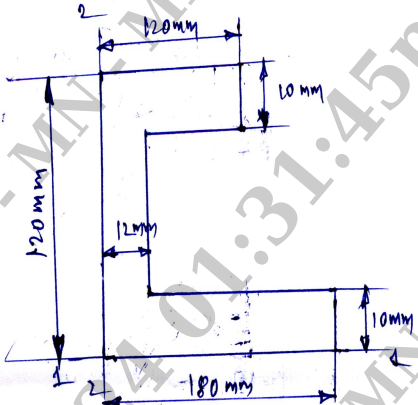
Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the scope of the following branches of Civil Engineering Disciplines : (i) Structural Engineering (ii) Surveying	10	L1	CO1
	b.	Explain the following building materials along with their application in construction, (i) Cement and Cement Mortar (ii) Pre-stressed Concrete.	10	L1	CO1
OR					
Q.2	a.	Explain the scope of the following branches of civil engineering disciplines: (i) Environmental engineering. (ii) Water resource and Irrigation engineering	10	L2	CO1
	b.	Explain the following structural elements of a building with their functions : (i) Masonry wall (ii) Staircase	10	L1	CO1
Module – 2					
Q.3	a.	Define sustainable development goals. List and explain the various goals of sustainable development set by united nations.	10	L2	CO2
	b.	Explain the cause of urban floods. List the remedial measures to control urban floods.	10	L2	CO2
OR					
Q.4	a.	Explain the methods of controlling sound and temperature to create a conclusive atmosphere in building.	10	L2	CO2
	b.	Explain different methods of managing the solid waste in urban areas.	10	L2	CO2
Module – 3					
Q.5	a.	State and prove parallelogram law of forces.	10	L1	CO3
	b.	The resultant of 2 forces, one of which is double the other is 260 N. If the direction of the larger force is reversed and the other remains unaltered, the resultant reduces to 180 N. Determine the magnitude of the forces and angle between forces.	10	L2	CO3

OR					
Q.6	a.	 Fig. Q6 (a) 2 kN force is the resultant of a system of forces acting along positive y-axis as shown in Fig. Q6 (a). Determine the value of F and θ.	10	L2	CO3
	b.	State and prove Varignon's theorem of moments.	10	L1	CO3
Module – 4					
Q.7	a.	Find the centroid of a semicircle.	10	L1	CO4
	b.	Determine the centroid of the Fig. Q7 (b).  Fig. Q7 (b)	10	L2	CO4
OR					
Q.8	a.	Determine the centroid of Fig. Q8 (a).  Fig. Q8 (a)	10	L2	CO4
	b.	Find the centroid of a triangle.	10	L1	CO4
Module – 5					
Q.9	a.	State and prove perpendicular axis and parallel axis theorem.	10	L1	CO5

	b.	Determine the moment of inertia of the unequal I-section about its centroidal axes as shown in Fig. Q9 (b).	10	L2	CO5
					
Fig. Q9 (b)					
OR					
Q.10	a.	Derive the moment of inertia of a triangle.	8	L2	CO5
	b.	Determine the moment of inertia of the section, shown in Fig. Q10 (b) about its centroidal axes. Calculate the least radius of gyration for the section.	12	L2	CO5
					
Fig. Q10 (b)					
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BETCK105B/BETCB105

First Semester B.E./B.Tech. Degree Examination, June/July 2024

Green Buildings

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is stabilized mud blocks? List out the advantages and disadvantages of stabilized mud block.	10	L1	CO1
	b.	Explain fibre reinforced polymer composites.	10	L2	CO1
OR					
Q.2	a.	What is concrete block? List out the advantages and disadvantages of concrete	10	L2	CO1
	b.	Explain the properties and advantages of Bamboo as building material.	10	L2	CO1
Module – 2					
Q.3	a.	Explain Rat trap bond with neat sketch.	10	L2	CO2
	b.	Explain filler slab and list out the advantages and disadvantages of filler slab.	10	L2	CO2
OR					
Q.4	a.	Explain Arches with neat sketch.	10	L2	CO2
	b.	Explain the different types of roofing systems.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the causes of Global warming.	10	L2	CO3
	b.	Briefly explain the global efforts to reduce carbon emission.	10	L2	CO3
OR					
Q.6	a.	Explain the environmental and economic benefits of green building.	10	L2	CO3
	b.	Briefly explain the features and necessity of green building.	10	L2	CO3
Module – 4					
Q.7	a.	Write short notes on the following : i) BREEAM ii) LEED.	10	L1	CO4
	b.	Explain Green design.	10	L2	CO4
OR					
Q.8	a.	Write a short note on the following i) GRIHA ii) Characteristics of sustainable building.	10	L1	CO4
	b.	Explain Integrated life cycle design of materials.	10	L2	CO4
Module – 5					
Q.9	a.	Write about the types of solar energy technologies.	10	L1	CO5
	b.	Explain various low energy approaches to water management.	10	L2	CO5
OR					
Q.10	a.	Write about utility of solar energy in buildings.	10	L1	CO5
	b.	Explain the management of sullage water and sewage.	10	L2	CO5

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Cell? Explain Prokaryotic cells and Eukaryotic cell in brief.	10	L2	CO1
	b.	With a neat figure, explain the structure of plant cell and animal cell.	10	L2	CO1
OR					
Q.2	a.	What is a biomolecule? Explain the classification of biomolecule.	10	L1	CO1
	b.	Explain the properties and functions of Enzymes.	10	L2	CO1
Module – 2					
Q.3	a.	What are Bioplastics? Compare the properties of PHA and PLA as bioplastics.	10	L1	CO1
	b.	With an example, explain the development of DNA vaccines.	10	L3	CO2
OR					
Q.4	a.	Explain the importance of lipids and its application in cleaning agents.	10	L2	CO1
	b.	Discuss the development of vaccine for Covid 19.	10	L3	CO2
Module – 3					
Q.5	a.	Discuss about the Human Brain as a CPU system.	10	L3	CO2
	b.	What is ECG? Describe the various parts of ECG.	10	L2	CO1
OR					
Q.6	a.	Describe the kidney as a filtration system.	10	L3	CO2
	b.	What are Pacemakers? Briefly explain the various kinds of pacemakers.	10	L2	CO2
Module – 4					
Q.7	a.	What is Echolocation? Discuss the application of echolocation in Ultrasonography.	10	L1	CO2
	b.	Explain the structure and design of Kingfisher's beak led to bullet trains.	10	L3	CO2
OR					
Q.8	a.	Discuss about the human blood substitutes.	10	L1	CO1
	b.	What is a lotus effect? Explain the mechanism and applications of lotus leaf effect.	10	L4	CO2

Module – 5					
Q.9	a.	Explain the architecture of Muscular and Skeletal systems as Scaffolds.	10	L3	CO4
	b.	Discuss the applications of Artificial intelligence in the diagnosis of disease.	10	L4	CO4
OR					
Q.10	a.	What is Bioprinting? Discuss the process and applications of Bio printing.	10	L4	CO4
	b.	Write a note on : (i) Electrical Tongue (ii) Self-healing Bioconcrete.	10	L3	CO4

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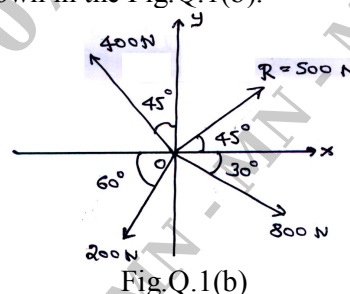
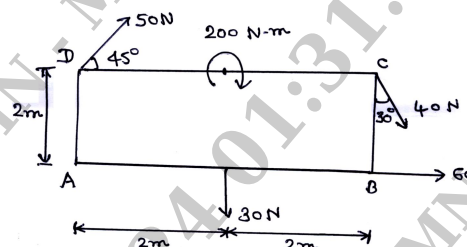
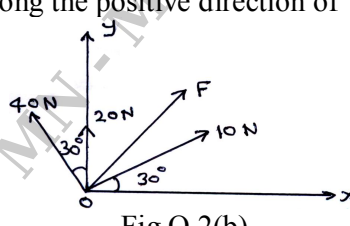
BCIVC103/203

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Engineering Mechanics

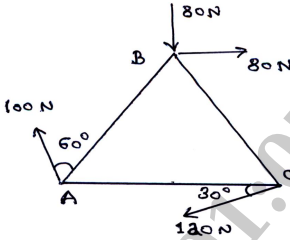
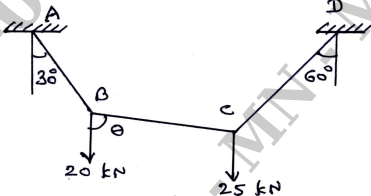
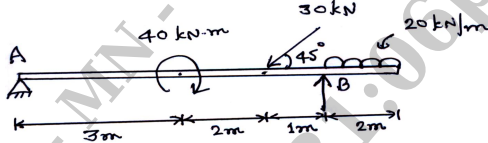
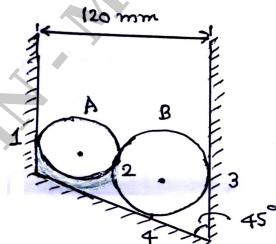
Time: 3 hrs.

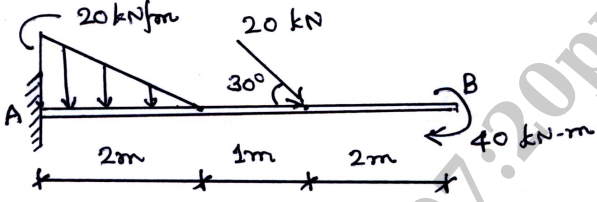
Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

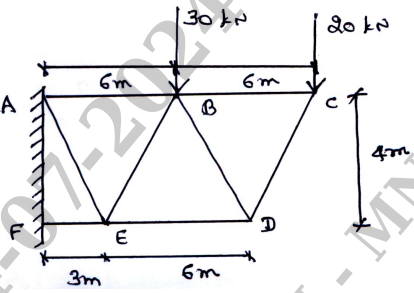
Module – 1				M	L	C
Q.1	a.	Explain the following: i) Principle of transmissibility of a force ii) Composition of forces and resolution of a force.		6	L2	CO1
	b.	Determine the fourth unknown force in magnitude and direction so that the resultant $\vec{R}$ acts as shown in the Fig.Q.1(b).  Fig.Q.1(b)		6	L3	CO1
	c.	Compute the resultant of the force system shown in the Fig.Q.1(c) with respect to point A. Also, locate the point where the resultant cuts the line AB.  Fig.Q.1(c)		8	L3	CO1, 2
OR						
Q.2	a.	State and prove principle of moments.		6	L2	CO1
	b.	Determine the unknown force $\vec{F}$ and its direction so that the resultant $\vec{R}$ of magnitude 72N acts along the positive direction of Y axis ($\uparrow$).  Fig.Q.2(b)		6	L3	CO1

1 of 5

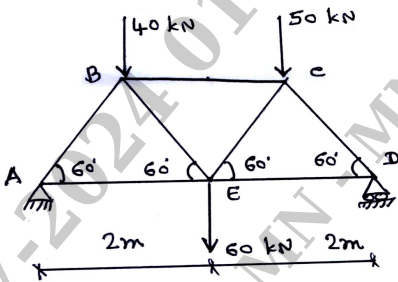
	c.	Compute the magnitude and direction of the resultant of the force system shown in the Fig.Q.2(c) with respect to point A of the equilateral triangle ABC. Side of triangle is 100mm. Also, find the location of the resultant along the edge AC.  Fig.Q.2(c)	8	L3	CO1
Module – 2					
Q.3	a.	Define equilibrium. State the conditions for the equilibrium of coplanar i) Concurrent force system ii) non-concurrent force system.	5	L2	CO2
	b.	In the given string system, determine the tensions in the strings and the angle θ for equilibrium.  Fig.Q.3(b)	7	L3	CO2
	c.	Determine the reactions in the beam shown in the Fig.Q.3(c).  Fig.Q.3(c)	8	L3	CO2
OR					
Q.4	a.	Distinguish between : i) Statically determinate and indeterminate beams. ii) Hinged support and fixed support.	6	L2	CO2
	b.	Compute the reactions at the contact points in the system shown (1, 2, 3, 4). Weight of sphere A = 50N Weight of sphere B = 80N Diameter of sphere A = 50mm Diameter of sphere B = 100mm.  Fig.Q.4(b)	8	L3	CO2

	c. Determine the support reactions in the beam shown :  Fig.Q.4(c)	6	L3	CO2
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Module – 3

Q.5	a. Determine the forces in the members of the truss shown in the figure by the method of joints.  Fig.Q.5(a)	10	L3	CO3
	b. State the laws of dry friction.	3	L2	CO3
	c. A weight 500N just starts moving down a rough inclined plane supported by a force of 200N acting parallel to the plane and it is at the point of moving up the plane when pulled by a force of 300N parallel to the plane. Find the inclination of the plane and the coefficient of friction between the inclined plane and the weight.	7	L3	CO3

OR

Q.6	a. Compute the forces in the members of the truss shown in the Fig.Q.6(a) by the method of joints.  Fig.Q.6(a)	10	L3	CO3
	b. Distinguish between angle of friction and angle of repose. Illustrate with a sketch.	3	L2	CO3
	c. A uniform ladder 4m long weighing 300N is placed against a vertical wall with an angle 60° with the floor. The coefficient of friction between the wall and the ladder is 0.25 and that between floor and ladder is 0.35. The ladder has to support a load of 1500N at its top. Find the horizontal force P to be applied at the bottom of the ladder to just prevent slipping.	7	L3	CO3

Module – 4

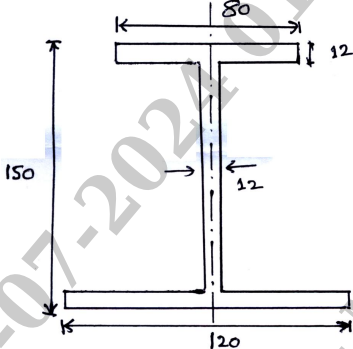
Q.7	a.	From first principles, derive the expression for locating the centroid of a semi-circular section.	6	L3	CO4
	b.	Illustrate: i) Parallel axis theorem ii) Perpendicular axis theorem.	4	L2	CO4
	c.	Determine the polar moment of inertia of the I-section shown in Fig.Q.7(c). All the dimensions are in mm. 	10	L3	CO4

Fig.Q.7(c)

OR

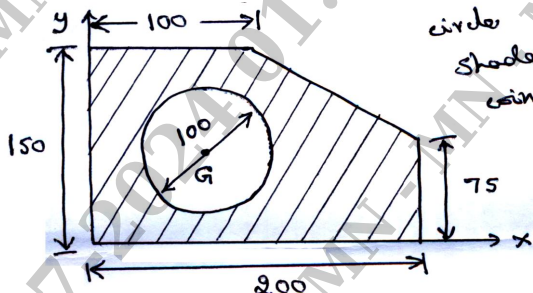
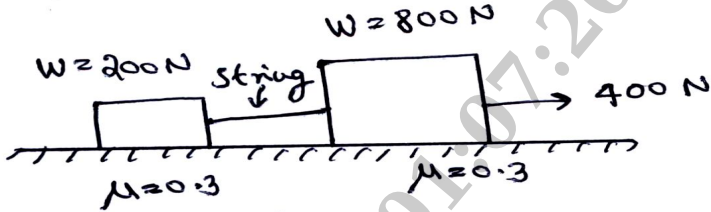
Q.8	a.	Derive the expression for the moment of inertia of a triangular section about its base. Hence, arrive at the expression about its parallel centroidal axis.	6	L3	CO4
	b.	Define and give the mathematical expressions for : i) Moment of inertia ii) Radius of gyration.	4	L2	CO4
	c.	Locate the centroid of the shaded lamina shown in the Fig.Q.8(c). Given that the centroid of the circle and the shaded lamina coincide. 	10	L3	CO4

Fig.Q.8(c)

Module – 5

Q.9	a.	Derive the three fundamental equations of linear motion.	6	L2	CO1
	b.	Determine the least initial velocity with which a projectile is to be projected so that it clears a wall 4m height located at a distance of 5m, and strikes the horizontal plane through the foot of the wall at a distance 4m beyond the wall. The point of projection is at the same level as the foot of the wall.	6	L3	CO5

	c.	Compute the acceleration of the system and the tension in the string shown in the Fig.Q.9(c). Adopt D'Alembert's principle.	8	L3	CO5
 Fig.Q.9(c)					
OR					
Q.10	a.	State and explain D'Alembert's principle. Give an example.	6	L2	CO5
	b.	A ball is thrown vertically upwards with an initial velocity of 36m/s. After 2 seconds, another ball is thrown vertically upwards. What should be its initial velocity so that it crosses first ball at a height of 30m?	8	L3	CO5
	c.	A projectile is aimed at a target on the horizontal plane and falls 12m short when the angle of projection is 15°, while it overshoots by 24m when the angle is 45°. Determine the angle of projection to hit the target.	6	L3	CO5

Third Semester B.E./B.Tech. Degree Examination, June/July 2024

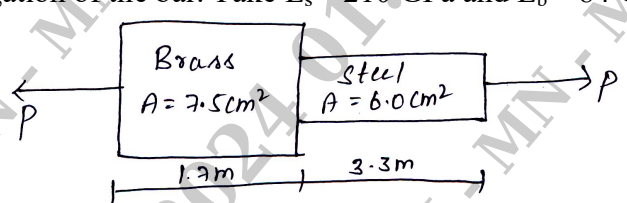
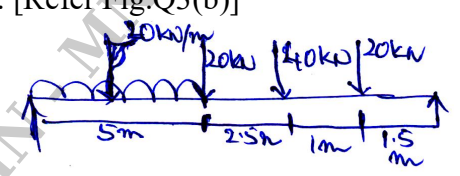
Strength of Materials

Time: 3 hrs.

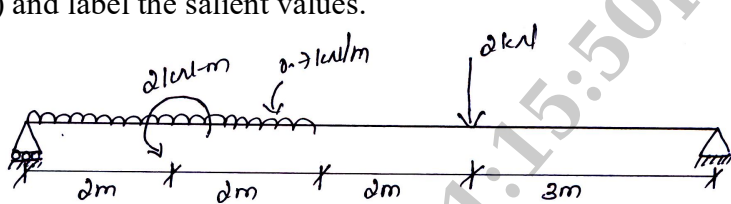
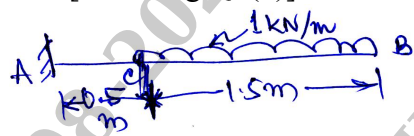
Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C												
Q.1	a.	A rod of length ‘L’ tapers uniformly from a diameter ‘D’ at one end to a diameter ‘d’ at the other. Derive the expression for the extension caused by an axial load ‘P’ the material has Young’s modulus of Elasticity ‘E’.	10	L3	CO1												
	b.	A specimen of 15mm diameter and 200mm long is subjected to tensile test and data at proportional and elastic limits were recorded as below. <table><tr><td>Limit</td><td>Stress (MPa)</td><td>Increase in length (mm)</td><td>Reduction in diameter (mm)</td></tr><tr><td>Proportional</td><td>340</td><td>0.90</td><td>?</td></tr><tr><td>Elastic</td><td>350</td><td>1.00</td><td>0.0225</td></tr></table> Find modulus of Elasticity, Poisson’s ratio and the reduction in diameter.	Limit	Stress (MPa)	Increase in length (mm)	Reduction in diameter (mm)	Proportional	340	0.90	?	Elastic	350	1.00	0.0225	10	L3	CO1
Limit	Stress (MPa)	Increase in length (mm)	Reduction in diameter (mm)														
Proportional	340	0.90	?														
Elastic	350	1.00	0.0225														
OR																	
Q.2	a.	A steel bar is 4 m long and its both ends are firmly fixed to two walls. The original temperature of the bar is 40°C. If the bar is cooled to 25°C, determine the thermal strain and stress in the bar. Assume $E_s = 200 \text{ kN/mm}^2$ and $\alpha_s = 12 \times 10^{-6}$ per °C. State the nature of stress set up.	10	L3	CO1												
	b.	A structural member 5m long is made up of two materials as shown in Fig.Q2(b). The bar is in tension under load ‘P’ and the total Elongation of the bar is 0.1 cm. Determine (i) The magnitude of the load (ii) The work done is Elongation of the bar. Take $E_s = 210 \text{ GPa}$ and $E_b = 84 \text{ GPa}$.  Fig.Q2(b)	10	L3	CO1												
Module – 2																	
Q.3	a.	Derive the relationship between Shear Force and Bending Moment.	05	L3	CO2												
	b.	A Beam of length 10m is simply supported at its ends. It carries a UDL of 20 kN/m run over the length of left half of its span together with concentrated loads of 20 kN, 40 kN and 20 kN situated at 1.5m, 2.5m and 5m respectively from right hand support. Draw the bending moment and shear force diagram. [Refer Fig.Q3(b)]  Fig.Q3(b)	15	L3	CO2												

OR

Q.4	a.	Draw Shear force and Bending moment diagram for the beam shown in Fig.Q4(a) and label the salient values.	12	L3	CO2
		 Fig.Q4(a)			
	b.	A cantilever beam of length 2.0m carries a UDL of 1 kN/m run over a length of 1.5m from free end. Draw the shear force and bending moment diagram for the cantilever. [Refer Fig.Q4(b)]	08	L3	CO2
		 Fig.Q4(b)			
Module – 3					
Q.5	a.	Prove that in case of rectangular section of a beam, the maximum shear stress is 1.5 times average shear stress.	10	L3	CO3
	b.	A hollow shaft is to transmit 300 kW power at 80 rpm. If the shear stress is not to exceed 60 N/mm^2 and the internal diameter is 0.6 times the external diameter, find the external and internal diameter assuming that the minimum torque is 1.4 times the mean.	10	L3	CO3
OR					
Q.6	a.	Derive the equation of pure bending with usual notations.	10	L3	CO3
	b.	A simply supported wooden beam of span 1.3m having a cross-section 150mm wide and 250mm deep carries a point load 'W' at the centre. The permissible stress are 7 N/mm^2 in bending and 1 N/mm^2 in shearing. Calculate the safe load 'W'.	10	L3	CO3
Module – 4					
Q.7	a.	What assumptions are made in Euler's formula? Deduce a formula for the critical load of a column having both ends hinged.	10	L3	CO4
	b.	A beam of length 5m and of uniform rectangular section is simply supported at its ends. It carries a UDL of 9 kN/m run over the entire length. Calculate the width and depth of the beam if the permissible bending stress is 7 N/mm^2 and the central deflection is not to exceed 1 cm.	10	L4	CO4
OR					
Q.8	a.	For a simply supported beam carrying a point load at the centre. Determine the magnitude of maximum deflection.	10	L3	CO4
	b.	A hollow mild steel tube 6m long and 4 cm internal diameter and 6mm thick is used as a strut. Find the crippling load and safe load taking FoS as 3. Take $E = 2 \times 10^5 \text{ N/mm}^2$ when (i) Both ends Hinged (ii) Both Ends Fixed.	10	L3	CO4
Module – 5					
Q.9	a.	Differentiate between thick and thin cylinder.	04	L2	CO5
	b.	Pressure inside a thin cylinder is 2115 Pa and its diameter 1 m. If thickness of the cylinder wall is 5mm, determine the hoop stress and longitudinal stress induced in the cylinder material.	06	L3	CO5
	c.	The tensile stresses at a point across two mutually perpendicular planes are 120 N/mm^2 and 60 N/mm^2 . Determine the normal, tangential and resultant stresses on a plane inclined at 30° to the axis of minor stress.	10	L3	CO5

OR

Q.10	a.	A point in a strained material possesses principal stresses of 600 N/mm^2 (tensile) and 400 N/mm^2 (compressive). Draw Mohr's stress circle and determine the following on an oblique plane, inclined at 40° with the plane of the major principal stress: i) Normal stress ii) Shear stress iii) Maximum shear stress iv) Resultant stress	10	L3	CO5
	b.	The external and internal diameter of a thick cylinder are respectively 800mm and 400mm. The cylinder is subjected to an external and internal fluid pressure of 100 GPa and 10 GPa. Determine the maximum hoop stress induced in the shell material.	10	L3	CO5

BCV302					
	b.	Explain temporary adjustment of dumpy level.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the characteristics of contours.	8	L1	CO3
	b.	What do you mean by contour? Explain the factors governing the choice of the proper contour interval.	8	L2	CO3
	c.	Differentiate between direct and indirect methods of contouring.	4	L2	CO3
OR					
Q.6	a.	Explain the procedure of data refinement and plotting in CAD using total station.	8	L2	CO3
	b.	Define the following: Station, Turning point, Fore sight, Back sight.	8	L1	CO3
	c.	With a neat sketch, explain profile leveling.	4	L2	CO3
Module – 4					
Q.7	a.	A railway embankment is 10m wide with side slope 1.5 to 1. Assuming ground to be level in a direction transverse to the centre line. Calculate the volume by prismoidal and trapezoidal formula. Contained in the length of 120m, the centre heights at 20m intervals being in meters 2.2, 3.7, 3.8, 4.0, 3.8, 2.8, 2.5.	10	L3	CO4
	b.	Obtain an expression for simple curve by Rankine's method.	10	L3	CO4
OR					
Q.8	a.	Two tangents intersect at a chainage of 1000 meter, the deflection angle being 28°. Calculate all the data necessary to set out a curve of 250 meter radius by Rankine's method and tabulate the results. Peg interval = 20m, least count of instrument = 20".	10	L3	CO4
	b.	The following perpendiculars offsets were taken at 10m intervals from a survey line to an irregular boundary line: 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20 and 5.65. Calculate area by trapezoidal and Simpson's rule.	10	L3	CO4
Module – 5					
Q.9	a.	List the GPS errors.	5	L1	CO5
	b.	What are the applications of Drones? Explain any one.	7	L2	CO5
	c.	Explain any two applications of remote sensing and GIS in Civil Engineering.	8	L2	CO5
OR					
Q.10	a.	List the different types of drones.	5	L1	CO5
	b.	What are the advantages of drones? Explain any one.	7	L2	CO5
	c.	Explain any four drone surveying requirements.	8	L2	CO5

CBCS SCHEME

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BCV303

Third Semester B.E/B.Tech. Degree Examination, June/July 2024 Engineering Geology

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C
1	a.	With neat sketch, explain internal structure of earth based on different discontinuities and add a note on its composition.	10	L2	CO2
	b.	Discuss the following terms : i) Focus and Epicenter ii) P and S waves.	4	L1	CO2
	c.	Define plate tectonics. Describe plate boundaries with their land forms.	6	L2	CO1
OR					
2	a.	Define land slide. Discuss its impact and preventive measures.	10	L2	CO1
	b.	Explain seismic zones of India as per Indian standard code and give safety measures on seismic resistant structure.	10	L3	CO2
Module – 2					
3	a.	Explain requirement of good building stones with suitable rock examples.	4	L2	CO3
	b.	Define mineral and rock. Explain 'light dependent' properties of minerals with an example.	8	L2	CO3
	c.	Differentiate between texture and structure in rocks and discuss textures in igneous rocks.	8	L3	CO2
OR					
4	a.	Briefly explain how igneous, sedimentary and metamorphic rocks are formed and enumerate their broad classification with suitable example.	6	L1	CO2
	b.	Discuss physical/geological and engineering properties along with their uses as building material for the following : i) Granite ii) Quartzite.	6	L3	CO3
	c.	Write the physical properties and industrial uses of two minerals each in the following group : i) Carbonate ii) Oxide.	8	L2	CO3
Module – 3					
5	a.	Define soil and its formation. Explain soil profile.	6	L2	CO2
	b.	Discuss chemical weathering and add a note on effect of weathering on monumental rocks.	8	L2	CO2
	c.	Define insitu and drifted soil. Enumerate the types of residual soil.	6	L3	CO3
1 of 2					

OR

6	a.	With neat trilinear diagram explain soil texture and describe engineering classification of soil based on grain size.	10	L3	CO2
	b.	Explain mechanical and biological weathering.	10	L2	CO2

Module – 4

7	a.	Discuss the formation of folds and faults and a note on their engineering significance.	6	L2	CO2
	b.	Define dam and explain its types. Discuss selection of dam site in folded and faulted rocks.	8	L3	CO3
	c.	A bed of limestone has a maximum dip of 30° along $N15^\circ E$. Find the amount of apparent dip along $N30^\circ W$. State the strike.	6	L3	CO3

OR

8	a.	Define dip, strike and outcrop with neat sketch explain recognition of any two geological structures in field.	10	L3	CO2
	b.	Three boreholes are sunk at three points of an equilateral triangle whose sides are 240 meters each. P is West of Q and R is North of midpoint between PQ, Boreholes P, Q and R reach the upper surface of a rich coal seam at 30 m, 70 m and 90 m depths respectively. i) Determine the attitude (Dip and strike) of the coal seam ii) One more borehole is proposed at S exactly at the centre of the triangle. Determine at what depth the new borehole 'S' reaches the upper bedding plane of the coal seam.	10	L4	CO3

Module – 5

9	a.	Define Aquifer and explain its types.	6	L2	CO4
	b.	Explain electric resistivity method and its applications in civil engineering and add a note on components, accessories of resistivity meter.	10	L3	CO4
	c.	Give a brief account on occurrence of ground water in igneous rocks.	4	L2	CO4

OR

10	a.	Define permissibility and explain factors affecting permeability.	6	L2	CO4
	b.	Explain seismic method, its types and application in civil engineering projects.	10	L3	CO4
	c.	Explain any two sedimentary rocks suitable for ground water occurrence.	4	L2	CO4

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Water Supply and Waste Water Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C								
Q.1	a.	Discuss the need of protected Water Supply.	10	L1	CO1								
	b.	List the various types of Water demand. Explain any three only.	10	L1	CO1								
OR													
Q.2	a.	What are the factors affecting per capita demand? Explain in detail.	10	L1	CO1								
	b.	Calculate probable population in the year 1980 , 1990 and 2000 by using arithmetical increase method. <table border="1"><tr><td>1940</td><td>1950</td><td>1960</td><td>1970</td></tr><tr><td>8000</td><td>12000</td><td>17000</td><td>22500</td></tr></table>	1940	1950	1960	1970	8000	12000	17000	22500	10	L3	CO1
1940	1950	1960	1970										
8000	12000	17000	22500										
Module – 2													
Q.3	a.	Draw unit flow diagram of water treatment plant, explain each unit in brief.	10	L2	CO2								
	b.	Explain the theory of Sedimentation tank. What are the types of Sedimentation tank? Explain any one.	10	L1	CO2								
OR													
Q.4	a.	Draw the neat sketches of Rapid sand filters. Explain working and cleaning of filters.	10	L2	CO2								
	b.	Design the approximate dimensions of a set of rapid gravity filters for treating water required for a population of 50,000. The rate of water supply being 180/Liters/day/person. The filters are rated to work 5000 Lit/hr/Sqm.	10	L3	CO2								
Module – 3													
Q.5	a.	What is Disinfection? What are the minor methods of disinfection? Explain any two methods.	10	L1	CO3								
	b.	What is Chlorination? What are the types of chlorination? Explain any two methods.	10	L1	CO3								
OR													
Q.6	a.	What are the types of Sewerage System? Explain their suitability in detail.	10	L1	CO3								
	b.	Explain any two Waste water physical , chemical and biological characteristics in detail.	10	L1	CO3								

Module – 4					
Q.7	a.	Draw flow diagram of Municipal waste water treatment unit operations and explain each units.	10	L2	CO4
	b.	Explain the importance of screens and types of screens in the sewage treatment process.	10	L1	CO4
OR					
Q.8	a.	Explain the working of conventional Activated Sludge Process (ASP) with flow diagram.	10	L2	CO5
	b.	What is Suspended growth process? What are the examples of suspended growth process units? Explain any one.	10	L1	CO5
Module – 5					
Q.9	a.	Explain the constructional details of a conventional trickling filters , with a neat sketch.	10	L1	CO5
	b.	The sewage flows from a primary settling tank to a standard rate trickling filter at a rate of 5 million liter per day having a 5 – day BOD of 150mg/lit. Determine the depth and the volume of the filter, adopting a surface loading of 2500ℓ/m ² /day and an urgent loading of 165/g/m ³ /day. Also determine the efficiency of the filter unit, using NRC formula.	10	L3	CO5
OR					
Q.10	a.	Explain the Rotating biological contactors, with neat sketch.	10	L2	CO5
	b.	Write a short note on : i) Oxidation ditch ii) Stabilization Ponds.	10	L1	CO5

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Third Semester B.E./B.Tech Degree Examination, June/July 2024

Sustainable Design Concept for Building services

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is sustainability? Explain the three pillars of sustainability.	10	L2	CO1
	b.	Define carbon footprint. Explain the methods to achieve net zero in carbon off setting.	10	L1	CO1
OR					
Q.2	a.	Define green building. Explain the factors affecting the site selection for green building.	10	L1	CO1
	b.	What are the elements of climate and how theory influence on building.	10	L2	CO1
Module – 2					
Q.3	a.	What is thermal comfort? Explain mechanics of heat transfer, with an example.	10	L2	CO2
	b.	What is building Acoustics? Explain the factors affecting.	10	L3	CO2
OR					
Q.4	a.	Define Indoor Air Quality. Explain methods to improve Indoor Air Quality.	10	L2	CO2
	b.	How do you assess visual comfort? Explain the enhancement strategies for day lighting and artificial lighting.	10	L3	CO2
Module – 3					
Q.5	a.	State the concept rain water harvesting. Explain the various methods of rain water harvesting.	10	L2	CO3
	b.	Demonstrate need for energy efficiency. Explain energy efficient HVAC and lighting as per ECBC – 2017.	10	L3	CO3
OR					
Q.6	a.	What are the types of waste and mention its treatment methods.	10	L2	CO3
	b.	Explain construction and demolition of waste management.	10	L2	CO3
Module – 4					
Q.7	a.	List and mention the various alternative materials used in the green building.	10	L1	CO4
	b.	What is life cycle assessment? Explain its types.	10	L2	CO4
OR					
Q.8	a.	Mention different phases of green building project management.	10	L2	CO4
	b.	Explain the contribution of lean construction in sustainable design.	10	L1	CO4
Module – 5					
Q.9	a.	Describe with the help of flowchart the “LEED” certification procedure.	10	L3	CO5
	b.	Explain the various stages involved in GRIHA building rating system.	10	L1	CO5
OR					
Q.10	a.	Explain the different Green building rating systems.	10	L1	CO5
	b.	Describe the criteria adopted during IGBC certification.	10	L3	CO5

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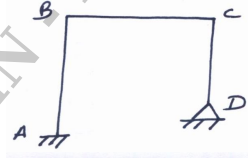
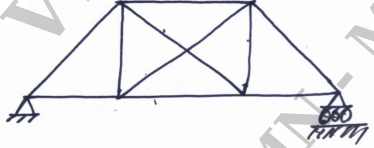
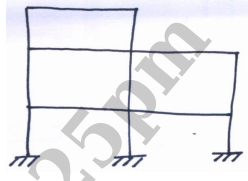
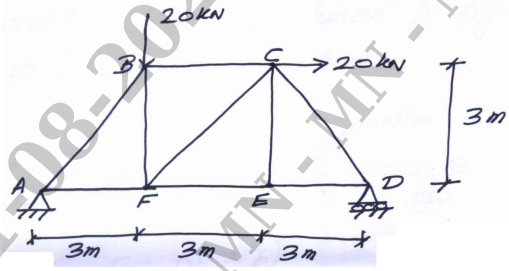
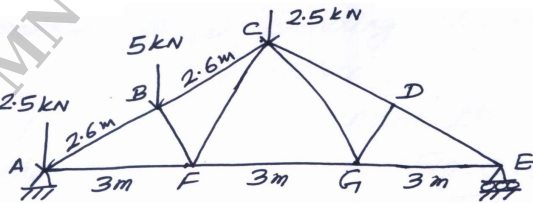
Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Analysis of Structures

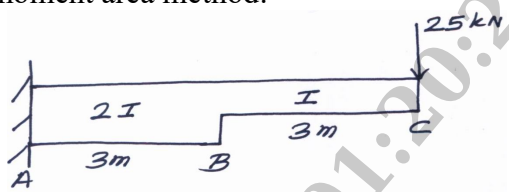
Time: 3 hrs.

Max. Marks: 100

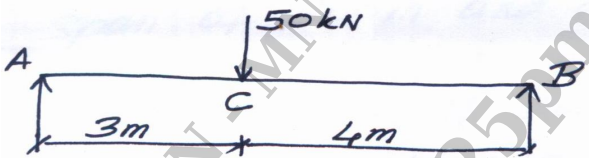
*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Differentiate between statically determinate and indeterminate beams with an example for each.		06	L1	CO1
	b.	Define degree of freedom. What is the degree of freedom for a : (i) Fixed support (ii) Hinged support		04	L1	CO1
	c.	Determine static and kinematic indeterminacy for the following structures shown in Fig.Q1(c). i)  ii)  iii)  iv)  v)  Fig.Q1(c)		10	L3	CO1
OR						
Q.2	a.	Determine the forces in all the members of the truss shown in Fig.Q2(a), adopting method of joints. 		10	L3	CO1
	b.	Determine the forces in all the members listed below using method of sections. [Refer Fig.Q2(b)] (i) Force in member CD (ii) Force in member CG (iii) Force in member FG. 		10	L3	CO1

Module – 2

Q.3	a.	Determine slope and deflection under the load for the beam as shown in Fig.Q3(a), using moment area method.	10	L3	CO2
		 Fig.Q3(a)			
	b.	Calculate the slope at support A and deflection at the centre of a simply supported beam as shown in Fig.Q3(b) by moment area method.	10	L3	CO2
		 Fig.Q3(b)			

OR

Q.4	a.	Obtain an expression for strain energy stored in a member when it is subjected to (i) Bending moment (ii) Shear	10	L2	CO2
	b.	Find the deflection under the load for the beam shown in Fig.Q4(b), by using Castiglino's theorem. Take $E = 2 \times 10^8 \text{ kN/m}^2$ and $I = 14 \times 10^{-6} \text{ m}^4$.	10	L3	CO2
		 Fig.Q4(b)			

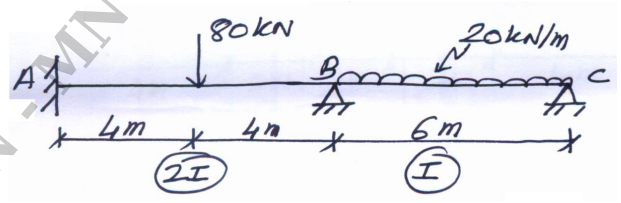
Module – 3

Q.5	a.	A symmetrical 3-hinged parabolic arch has a span of 25 m. It carries a UDL of intensity 20 kN/m over the entire span and 2 point loads of 20 kN each at 3 m and 6 m from the left support. Compute the reactions at the supports and also find the bending moment, radial shear and normal thrust at a section 4m from the left end. Take central rise as 5m.	12	L3	CO3
	b.	Show that the shape of parabolic arch is a funicular shape for a three hinged arch subjected to UDL over its entire span.	08	L2	CO3

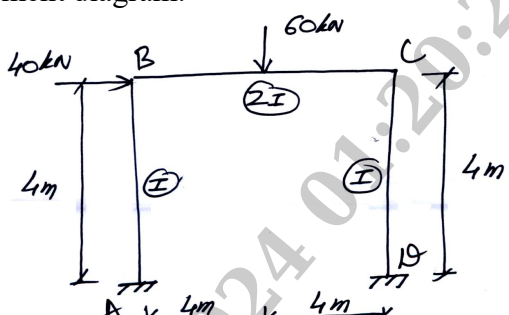
OR

Q.6		A symmetrical unstiffened suspension cable is parabolic in shape and has a span of 300 m and a dip of 30 m. It supports a UDL of 20 kN/m over the whole span. If the maximum allowable stress is 150 N/mm^2 , determine the length and area of the cable. What would be the increase in length and sag for a rise of temperature of 50°F . Given, coefficient of expansion, $\alpha = 6 \times 10^{-6} \text{ per } ^\circ\text{F}$.	20	L3	CO3
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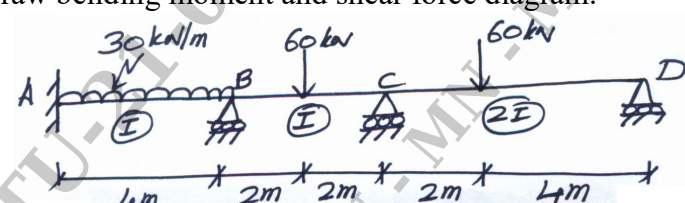
Module – 4

Q.7		Analyze the beam completely by slope deflection method, when support B sinks by 1 mm and support C rises by 0.5 mm. Take $EI = 30000 \text{ kN.m}^2$. Refer Fig.Q7. Draw BMD, SFD and Elastic Curve.	20	L3	CO4
		 Fig.Q7			

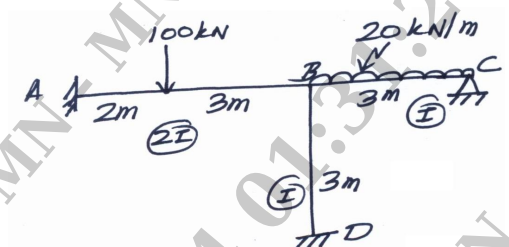
OR

Q.8	Analyze the portal frame shown in Fig.Q8 by slope-deflection method. Draw bending moment diagram.  Fig.Q8	20	L3	CO4
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Module – 5

Q.9	Analyze the continuous beam shown in Fig.Q9 by moment distribution method. Draw bending moment and shear force diagram.  Fig.Q9	20	L3	CO5
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OR

Q.10	Analyze the frame by moment distribution method. Draw BMD and SFD. Refer Fig.Q10.  Fig.Q10	20	L3	CO5
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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Fluid Mechanics and Hydraulics

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define the following terms along with symbols and units : (i) Compressibility (ii) Capillarity (iii) Surface tension	6	L1	CO1
	b.	State and prove Pascal's law.	6	L2	CO1
	c.	A tube differential manometer connects two pipes A and B. Pipe A contains carbon tetra chloride having specific gravity 1.594 under a pressure of 117.72 kN/m ² and pipe B contains oil of specific gravity 0.8 under a pressure of 117.72 kN/m ² . The pipe A lies 2.5 m above pipe B. Find the difference in pressure measured by mercury as fluid filling U-tube. Assume mercury in the right limb is 50 cm below centre of pipe B.	8	L2	CO1
OR					
Q.2	a.	Differentiate between : (i) Pressure intensity and Pressure head (ii) Simple and differential manometers. (iii) Absolute and Gauge pressure.	6	L2	CO1
	b.	Derive an expression for total pressure and centre of pressure on a plane surface immersed vertically in water.	6	L2	CO1
	c.	A 1200 mm × 1800 mm size rectangular plate is immersed in water with an inclination of 30° to the horizontal. The 1200 mm side of the plate is kept horizontal at a depth of 30 m below the water surface. Compute the total pressure on the surface and the position of centre of pressure.	8	L3	CO1
Module – 2					
Q.3	a.	Differentiate between : (i) Laminar and turbulent flow (ii) Uniform and non uniform flow (iii) Steady and Unsteady flow	6	L2	CO2
	b.	Derive continuity equation for a three dimensional flow in Cartesian co-ordinates.	6	L2	CO2
	c.	The velocity potential function is given by $\phi = 5(x^2 - y^2)$. Calculate the velocity components at the points (4, 5)	8	L3	CO2
OR					
Q.4	a.	List the assumptions made in deriving Bernoulli's equation.	6	L2	CO2
	b.	Derive the equation for the discharge through Venturimeter.	6	L2	CO2
	c.	Crude oil of $G = 0.84$ flow through a pipe with a rate of 450 lps. The diameter of pipe and pressure in the pipe at one section are respectively 25 cm and 55 kPa and at section two are 50 cm and 320 kPa. Find the direction of flow through the pipe and head loss. Pipe is horizontal.	8	L3	CO2

Module – 3					
Q.5	a.	Define Orifice and Mouth piece. Also derive the hydraulic co-efficients experimentally.	6	L2	CO3
	b.	Derive an expression for discharge over a triangular notch.	6	L2	CO3
	c.	A rectangular channel 2 m wide has a discharge 250 lps, which is measured by a right angled V-notch weir. Find the position of the apex of the notch from the bed of the channel of maximum depth of water is not to exceed 1.3 m. Take $C_d = 0.62$.	8	L3	CO3
OR					
Q.6	a.	Define : (i) Major and Minor losses in a pipe flow. (ii) Pipes in series and parallel (iii) Water hammer in pipe flow	6	L2	CO3
	b.	Derive Darcy-Weisbach equation for head loss due to friction in a pipe.	6	L2	CO3
	c.	Water is required to be supplied to a colony of 4000 residents at a rate of 180 litres per person from a source 3 km away. If half the daily requirement needs to be pumped in 8 hours against a friction of 18 m, find the size of the main pipe supplying water. Assume friction factor as 0.028.	8	L3	CO3
Module – 4					
Q.7	a.	Define most economical channel section. For the most economical trapezoidal section show that half of top width is equal to the side slope length.	6	L2	CO4
	b.	What is specific energy curve? Draw and derive expressions for critical depth and critical velocity.	6	L2	CO4
	c.	The discharge of water through a rectangular channel of width 8 m is $15 \text{ m}^3/\text{s}$. When the depth of flow of water is 1.2 m. Calculate (i) Specific energy of flowing water. (ii) Critical depth and critical velocity. (iii) Value of minimum specific energy.	8	L3	CO4
OR					
Q.8	a.	What is gradually varied flow? Derive the dynamic equation for gradually varied flow.	6	L2	CO4
	b.	Derive an expression for conjugate depths in case of hydraulic jump in a rectangular channel laid horizontal.	6	L2	CO4
	c.	A sluice gate discharges water into a horizontal rectangular channel with a velocity of 6 m/s and depth of flow is 0.4 m. The width of the channel is 8 m. Determine whether a hydraulic jump will occur, and if so, find its height and loss of energy per kg of water. Also determine the power lost in the hydraulic jump.	8	L3	CO4
Module – 5					
Q.9	a.	Obtain an expression for force exerted by a jet striking at the centre of a moving symmetrical curved vane. And show that its maximum efficiency is limited to $\frac{16}{27}$.	6	L3	CO5
	b.	Give the classification of turbines based on different criteria.	6	L2	CO5
	c.	A jet of water having velocity 45 m/s impinges without shock on a series of curved vanes moving at 15 m/s, the direction of motion of vanes being 20° to that of jet. The relative velocity at the outlet is 0.9 of that at inlet and the absolute velocity of water at the exit is to be normal to the motion of vanes. Find (i) Vane angles at entrance and exit. (ii) Hydraulic efficiency.	8	L3	CO5

OR					
Q.10	a.	By means of neat sketch, explain the Francis turbine.	6	L2	CO5
	b.	A Pelton wheel turbine has to be designed for the following : Data : Power = 6000 kW, Net head = 300 m, Speed = 550 rpm, Jet ratio = $\frac{1}{10}$, Overall efficiency = 85%, $C_v = 0.98$, Speed ratio = 0.46 Determine diameter of runner and jet, discharge and number of jets required.	6	L3	CO4
	c.	Explain various efficiencies of centrifugal pumps. Also define (i) Manometric head (ii) Static head (iii) Suction head (iv) Delivery head	8	L1	CO5

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Transportation Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Mention the different modes of transportation. Explain the characteristics of road transport.	6	L1	CO1
	b.	Mention the Jayakar committee recommendations and its implementations.	6	L1	CO1
	c.	Calculate the minimum sight distance required to avoid a head on collision of 2 cars approaching from the opposite direction at 90 and 60 kmph. Assume a reaction time of 2.5 sec, co-efficient of friction of 0.7 and break efficiency of 50% in both the cases.	8	L2	CO1
OR					
Q.2	a.	Explain Engineering surveys.	6	L1	CO1
	b.	Mention the various cross-sectional elements to be designed for a highway and explain them briefly.	6	L1	CO1
	c.	Vertical summit curve is formed due to the intersection of 2 gradients +5% and -6%. Design the length of summit curve to provide SSD for a design speed of 80kmph. Assume any other data suitably.	8	L2	CO1
Module – 2					
Q.3	a.	Explain the desirable properties of subgrade soil.	6	L2	CO2
	b.	With a neat sketch, explain the functions of component parts of flexible pavement.	6	L2	CO2
	c.	Explain the significance of Highway Drainage.	8	L2	CO2
OR					
Q.4	a.	Explain the desirable properties of road aggregates.	6	L2	CO2
	b.	Distinguish between flexible pavement and rigid pavement.	6	L2	CO2
	c.	List the objectives of i) Surface drainage ii) Sub-surface drainage of roads.	8	L2	CO2
Module – 3					
Q.5	a.	List the different road user characteristics and explain the concept of PIEV theory.	10	L2	CO3
1 of 2					

BCV403					
	b.	Following data were obtained from the spot speed studies suggest: i) Speed limit for regulation ii) Lower speed causing congestion iii) Speed to check the geometric design elements Speed range (kmph) No. of vehicles 5-10 230 10-15 375 15-20 500 20-25 680 25-30 525 30-35 430 35-40 290 40-50 110 50-60 25 60-70 8	10	L3	CO3
OR					
Q.6	a.	Discuss the various types of traffic studies. What are the objects of carrying out traffic volume studies?	10	L2	CO3
	b.	The average normal flow on cross roads A and B during design period are 400 and 250 PCU per hour. The saturation flows are 1250 and 1000 PCU per hour respectively. The all red time required for pedestrian crossing is 12 seconds. Design a two-phase signal by Webster's method.	10	L3	CO3
Module – 4					
Q.7	a.	Describe the requirements of an ideal permanent way.	10	L2	CO4
	b.	Determine the quantity of materials required to construct a 800m long B.G. railway track, assuming a sleeper, density of (n + 5).	10	L2	CO4
OR					
Q.8	a.	What are the functions and requirements of sleepers and Ballast?	10	L2	CO4
	b.	If a 8° curve track diverges from a main curve of 5° in an opposite direction in the layout of a B.G yard, calculate the super elevation and the speed on the branch line, if the maximum speed permitted on the main line is 45kmph.	10	L2	CO4
Module – 5					
Q.9	a.	List the various elements of an air port and explain with a neat sketch.	10	L2	CO4
	b.	Describe the elements of taxiway geometric design.	10	L2	CO4
OR					
Q.10	a.	Explain the factors, which influence the site selection for airport.	10	L2	CO4
	b.	List the assumed conditions under which basic runway length is determined. Explain the normal landing case.	10	L2	CO4

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Concreting Techniques and Practices

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. Use of code book IS 10262-2019 for mix design problem is permitted.*

Module – 1			M	L	C																
Q.1	a.	Mention different types of cement. Explain briefly.	10	L2	CO1																
	b.	Explain the following : i) Fly ash ii) Silica fumes iii) Rice husk ash iv) GGBS.	10	L2	CO1																
OR																					
Q.2	a.	Enumerate and explain the properties of aggregates for concrete.	10	L3	CO1																
	b.	Explain the various tests conducted on Recycled concrete aggregate.	10	L2	CO1																
Module – 2																					
Q.3	a.	Explain briefly : i) Super plasticizer ii) Accelerators	10	L2	CO2																
	b.	Explain the qualities of water used for manufacture of concrete specify the limits.	10	L2	CO2																
OR																					
Q.4	a.	Mention the necessity of aggregate blending and briefly explain any one method of blending.	10	L1	CO2																
	b.	Draw gradation curve and determine Fineness modulus of aggregate from following particular. <table border="1"><tr><td>Sieve size (mm)</td><td>25</td><td>19</td><td>12.5</td><td>9.5</td><td>4.75</td><td>2.36</td><td>Pan</td></tr><tr><td>Mass Retained (gms)</td><td>0</td><td>405</td><td>2850</td><td>2435</td><td>2030</td><td>375</td><td>35</td></tr></table>	Sieve size (mm)	25	19	12.5	9.5	4.75	2.36	Pan	Mass Retained (gms)	0	405	2850	2435	2030	375	35	10	L3	CO2
Sieve size (mm)	25	19	12.5	9.5	4.75	2.36	Pan														
Mass Retained (gms)	0	405	2850	2435	2030	375	35														
Module – 3																					
Q.5		Design a concrete mix for M25 a) Grade designation – M25 b) Type of cement – OPC43 c) Max Nominal size of Aggregate – 20mm d) Min cement content – 300Kg/m ³ e) Water cement Ratio – 0.50 f) Workability – 75mm slump g) Exposure condition – moderate (RCC) h) Max.cement content – 450Kg/m ³ i) Chemical admixture – Nil j) F.A zone – Zone 2 k) S.G of cement – 3.15 l) C.A : SG – 2.80 m) C.A : water absorption – 1% n) F.A : SG – 2.65 o) F.A : Water absorption – 2%	20	L3	CO3																

OR					
Q.6	a.	List and explain factors affecting workability.	10	L2	CO3
	b.	Define slump of concrete, explain procedure to determine slump of concrete using slump cone method.	10	L2	CO3
Module – 4					
Q.7	a.	With neat sketch, explain component of batching plant.	10	L1	CO4
	b.	What are the methods of transportation of concrete used to produce good concrete? Explain.	10	L2	CO4
OR					
Q.8	a.	Explain the different methods of concrete curing and define curing of concrete.	10	L2	CO4
	b.	Enumerate necessity of R.M.C with advantages and disadvantages.	10	L2	CO4
Module – 5					
Q.9	a.	List the types of fibres used in F.R.C and discuss the factors affecting properties of F.R.C	10	L2	CO5
	b.	What is S.C.C (Self Compacting Concrete]? Explain the properties of S.C.C.	10	L1	CO5
OR					
Q.10	a.	Write a note on : i) Geo polymer concrete ii) High performance concrete	10	L2	CO5
	b.	Mention the types of Shrinkage and factors affecting Shrinkage.	10	L2	CO5
